

Online Appendix: Nominal Rigidities and Asset Pricing

Michael Weber

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A. Model Derivations

A.1 Households

The representative household has additively separable utility in consumption and leisure and maximizes

$$\begin{aligned} & \mathbb{E}_t \sum_{s=0}^{\infty} \left[\beta^s \frac{(C_{t+s} - bC_{t+s-1})^{1-\gamma}}{1-\gamma} - \psi_L \int_0^1 \frac{h_{i,t+s}^{1+\sigma}}{1+\sigma} di \right] \\ \text{s.t.} \quad & P_t C_t = \int_0^1 W_{i,t} h_{i,t} di + R_{t-1} B_{t-1} - B_t + D_t, \end{aligned}$$

where $\mathbb{E}_t$ is the expectation operator conditional on the time t information set of the representative household, β is the discount factor, C is the composite consumption good, $b \geq 0$ is a habit persistence parameter in consumption, $h_{i,t}$ denotes hours worked of type i , $\psi_L \geq 0$ is a parameter, P is the composite price index defined below, $W_{i,t}$ time t nominal wage for labor type i , R_t is the gross nominal interest rate, B_t denotes nominal bond holdings, and D_t are aggregate dividends by the firm sector. Profits are redistributed via lump-sum transfer at the end of each period. The parameters γ and σ denote, respectively, the coefficient of relative risk aversion and the inverse of the Frisch elasticity of labor supply.

The first order conditions for the representative household for consumption and bond holdings are given by:

$$(C_t - bC_{t-1})^{-\gamma} = \Lambda_t P_t = \lambda_t$$

$$\mathbb{E}_t(\beta \Lambda_{t+1} R_t) = \Lambda_t.$$

Hence,

$$\begin{aligned} \frac{1}{P_t} (C_t - bC_{t-1})^{-\gamma} &= \beta R_t \mathbb{E}_t \left[\frac{1}{P_{t+1}} (C_{t+1} - bC_t)^{-\gamma} \right] \\ 1 &= \beta R_t \mathbb{E}_t \left[\frac{1}{\pi_{t+1}} \left(\frac{C_{t+1} - bC_t}{C_t - bC_{t-1}} \right)^{-\gamma} \right]. \end{aligned}$$

A.2 Optimal Consumption Allocation

The composite consumption good is given by:

$$C_t \equiv \left[\int_0^1 f(k)^{\frac{1}{\varepsilon_c}} C_{k,t}^{\frac{\varepsilon_c-1}{\varepsilon_c}} dk \right]^{\frac{\varepsilon_c}{\varepsilon_c-1}}, \quad (\text{A.1})$$

$$C_{k,t} \equiv f(k) \left[\int_0^1 C_{kj,t}^{\frac{\varepsilon_{ck}-1}{\varepsilon_{ck}}} dj \right]^{\frac{\varepsilon_{ck}}{\varepsilon_{ck}-1}}. \quad (\text{A.2})$$

Let $P_{kj,t}$ denote the price charged by firm j in sector k for the consumption good C_{kj} in period t . The consumption price index is then given by:

$$P_t = \left[\int_0^1 f(k) P_{k,t}^{1-\varepsilon_c} dk \right]^{\frac{1}{1-\varepsilon_c}} \quad (\text{A.3})$$

$$P_{k,t} = \left[\int_0^1 P_{kj,t}^{1-\varepsilon_{ck}} dj \right]^{\frac{1}{1-\varepsilon_{ck}}}. \quad (\text{A.4})$$

Assume that the representative household maximizes equation (A.2) for any given expenditure level Q_t :

$$\int_0^1 P_{kj,t} C_{kj,t} dj = Q_t. \quad (\text{A.5})$$

The Lagrangian associated with this problem is given by:

$$\mathfrak{L} = f(k) \left(\int_0^1 C_{kj,t}^{\frac{\varepsilon_{ck}-1}{\varepsilon_{ck}}} dj \right)^{\frac{\varepsilon_{ck}}{\varepsilon_{ck}-1}} - \Omega_t \left(\int_0^1 P_{kj,t} C_{kj,t} dj - Q_t \right),$$

where Ω_t is the associated Lagrange multiplier.

The first-order conditions are:

$$f(k) \frac{\varepsilon_{ck}}{\varepsilon_{ck}-1} \left(\int_0^1 C_{kj,t}^{\frac{\varepsilon_{ck}-1}{\varepsilon_{ck}}} dj \right)^{\frac{1}{\varepsilon_{ck}-1}} \frac{\varepsilon_{ck}-1}{\varepsilon_{ck}} C_{kj,t}^{-\frac{1}{\varepsilon_{ck}}} = \Omega_t P_{kj,t} \quad \forall j \in [0, 1].$$

Rearranging and dividing by the expression for subcomposite $C_{ki,t}$:

$$C_{kj,t} = C_{ki,t} \left(\frac{P_{kj,t}}{P_{ki,t}} \right)^{-\varepsilon_{ck}}.$$

Substituting this expression into equation (A.5) we get:

$$\int_0^1 P_{kj,t} C_{ki,t} \left(\frac{P_{kj,t}}{P_{ki,t}} \right)^{-\varepsilon_{ck}} dj = Q_t$$

$$\Longleftrightarrow$$

$$C_{ki,t} P_{ki,t}^{\varepsilon_{ck}} \int_0^1 P_{kj,t}^{1-\varepsilon_{ck}} dj = Q_t.$$

Substituting for $\int_0^1 P_{kj,t}^{1-\varepsilon_{ck}} dj$ using equation (A.4):

$$C_{ki,t} P_{ki,t}^{\varepsilon_{ck}} P_{k,t}^{1-\varepsilon_{ck}} = Q_t \quad (\text{A.6})$$

$$\Longleftrightarrow$$

$$C_{ki,t} = \frac{Q_t}{P_{k,t}} \left(\frac{P_{ki,t}}{P_{k,t}} \right)^{-\varepsilon_{ck}}.$$

Substituting the last expression for subcomposite kj in equation (A.2):

$$C_{k,t} = f(k) \left\{ \int_0^1 \left[\frac{Q_t}{P_{k,t}} \left(\frac{P_{kj,t}}{P_{k,t}} \right)^{-\varepsilon_{ck}} \right]^{\frac{\varepsilon_{ck}-1}{\varepsilon_{ck}}} dj \right\}^{\frac{\varepsilon_{ck}}{\varepsilon_{ck}-1}}.$$

Using the definition of the price index, equation (A.4), we get:

$$\begin{aligned} C_{k,t} &= f(k) \frac{Q_t}{P_{k,t}} \\ \iff \\ \frac{Q_t}{P_{k,t}} &= C_{k,t} f(k)^{-1}. \end{aligned} \tag{A.7}$$

Combining equation (A.7) with equation (A.6) for good kj :

$$C_{k,t} f(k)^{-1} = C_{kj,t} \left(\frac{P_{kj,t}}{P_{k,t}} \right)^{\varepsilon_{ck}}.$$

Solving for $C_{kj,t}$ we arrive at the demand for subcomposite kj :

$$C_{kj,t} = f(k)^{-1} C_{k,t} \left(\frac{P_{kj,t}}{P_{k,t}} \right)^{-\varepsilon_{ck}}.$$

Following the same logic as above, we can derive the demand for the composite consumption good for a given expenditure level of Z_t :

$$\int_0^1 P_{k,t} C_{k,t} dk = Z_t. \tag{A.8}$$

The Lagrangian is given by:

$$\mathcal{L} = \left(\int_0^1 f(k)^{\frac{1}{\varepsilon_c}} C_{k,t}^{\frac{\varepsilon_c-1}{\varepsilon_c}} dk \right)^{\frac{\varepsilon_c}{\varepsilon_c-1}} - \Theta_t \left(\int_0^1 P_{k,t} C_{k,t} dk - Z_t \right),$$

where Θ_t is the associated Lagrange multiplier. The first-order conditions are:

$$\frac{\varepsilon_c}{\varepsilon_c - 1} \left(\int_0^1 f(k)^{\frac{1}{\varepsilon_c}} C_{k,t}^{\frac{\varepsilon_c-1}{\varepsilon_c}} dk \right)^{\frac{1}{\varepsilon_c-1}} \frac{\varepsilon_c - 1}{\varepsilon_c} C_{k,t}^{-\frac{1}{\varepsilon_c}} f(k)^{\frac{1}{\varepsilon_c}} = \Theta_t P_{k,t} \quad \forall k \in [0, 1].$$

Rearranging and dividing by the expression for the consumption composite of sector 1,

$C_{kl,t}$:

$$C_{k,t} = C_{l,t} \frac{f(k)}{f(l)} \left(\frac{P_{k,t}}{P_{l,t}} \right)^{-\varepsilon_c}.$$

Substituting this expression into equation (A.8) we get:

$$\int_0^1 P_{k,t} \frac{f(k)}{f(l)} C_{l,t} \left(\frac{P_{k,t}}{P_{l,t}} \right)^{-\varepsilon_c} dk = Z_t$$

$\iff$

$$C_{l,t} P_{l,t}^{\varepsilon_c} \frac{1}{f(l)} \int_0^1 f(k) P_{k,t}^{1-\varepsilon_c} dk = Z_t.$$

Substituting for $\int_0^1 f(k)P_{k,t}^{1-\varepsilon_c}dk$ using equation (A.3)

$$C_{l,t}P_{l,t}^{\varepsilon_c}\frac{1}{f(l)}P_t^{1-\varepsilon_c} = Z_t \quad (\text{A.9})$$

$$\Longleftrightarrow$$

$$C_{l,t} = f(l)\frac{Z_t}{P_t}\left(\frac{P_{l,t}}{P_t}\right)^{-\varepsilon_c}.$$

Substituting the last expression for the sector k consumption good in equation (A.1):

$$C_t = \left\{ \int_0^1 f(k)^{\frac{1}{\varepsilon_c}} \left[f(k)\frac{Z_t}{P_t}\left(\frac{P_{k,t}}{P_t}\right)^{-\varepsilon_c} \right]^{\frac{\varepsilon_c-1}{\varepsilon_c}} dk \right\}^{\frac{\varepsilon_c}{\varepsilon_c-1}}.$$

Using the definition of the price index, equation (A.3), we get:

$$C_t = \frac{Z_t}{P_t}. \quad (\text{A.10})$$

Combining equation (A.10) with equation (A.9) for sector k :

$$\frac{Z_t}{P_t} = \frac{1}{f(k)}C_{k,t}\left(\frac{P_{k,t}}{P_t}\right)^{\varepsilon_c}.$$

Solving for $C_{k,t}$ we arrive at the demand for sector k composite consumption good:

$$C_{k,t} = f(k)C_t\left(\frac{P_{k,t}}{P_t}\right)^{-\varepsilon_c}.$$

A.3 Demand for Different Labor Types

The competitive labor contractor demands labor of different types i to maximize:

$$\begin{aligned} & W_t H_t - \int_0^1 W_{i,t} h_{i,t} di \\ \text{s.t.} \quad & H_t = \left[\int_0^1 h_{i,t}^{\frac{\varepsilon_w-1}{\varepsilon_w}} di \right]^{\frac{\varepsilon_w}{\varepsilon_w-1}}, \end{aligned}$$

where W is the aggregate wage rate, H is homogeneous labor, W_i is wage rate of labor type i , and h_i and $\varepsilon_w \geq 1$ is the elasticity of substitution among different labor types.

Following similar steps as in the derivation for the optimal consumption allocation, we arrive at:

$$h_{i,t} = \left(\frac{W_{i,t}}{W_t}\right)^{-\varepsilon_w} H_t.$$

The resulting demand curves are identical to a setup without competitive labor contractor where firms directly demand different labor types.

Note that the aggregate wage rate, W can be simply derived from:

$$\begin{aligned} W_t H_t &= \int_0^1 W_{i,t} h_{i,t} di \\ &= \int_0^1 W_{i,t} \left(\frac{W_{i,t}}{W_t} \right)^{-\varepsilon_w} H_t di \\ W_t^{1-\varepsilon_w} &= \int_0^1 W_{i,t}^{1-\varepsilon_w} di. \end{aligned}$$

A.4 Optimal Reset Wage

Let U_t be the optimal reset wage. The optimizing problem of the union representing labor of type i, $h_{i,t}$, is given by:

$$\begin{aligned} \mathbb{E}_t \sum_{s=0}^{\infty} (\beta \theta_w)^s \left\{ -\psi_L \frac{h_{i,t+s}^{1+\sigma}}{1+\sigma} + \frac{\Lambda_{t+s}}{\Lambda_t} U_{i,t} h_{i,t+s} \right\} \\ \text{s.t.} \quad h_{i,t+s} = \left(\frac{U_{i,t}}{W_{t+s}} \right)^{-\varepsilon_w} H_{t+s}, \end{aligned}$$

where θ_w is the probability that the union cannot reset the wage rate of labor type i, $W_{i,t}$; σ is the inverse of the Frisch labor supply elasticity; Λ is the Lagrange multiplier on the households budget constraint; ε_w is the elasticity of substitution among different labor types; and ψ_L is a parameter.

The first order condition is given by:

$$\begin{aligned} \mathbb{E}_t \sum_{s=0}^{\infty} (\beta \theta_w)^s \left\{ -\psi_L h_{i,t+s}^{1+\sigma} \frac{1}{U_{i,t}} (-\varepsilon_w) + \Lambda_{t+s} (1 - \varepsilon_w) h_{i,t+s} \right\} &= 0 \\ \iff \\ \mathbb{E}_t \sum_{s=0}^{\infty} (\beta \theta_w)^s \left\{ \lambda_{t+s} \left(\frac{U_{i,t}}{P_{t+s}} \right) \left(\frac{U_{i,t}}{W_{t+s}} \right)^{-\varepsilon_w} H_{t+s} - \frac{\varepsilon_w}{\varepsilon_w - 1} \psi_L \left(\frac{U_{i,t}}{W_{t+s}} \right)^{-\varepsilon_w(1+\sigma)} H_{t+s}^{1+\sigma} \right\} &= 0 \\ \iff \\ \mathbb{E}_t \sum_{s=0}^{\infty} (\beta \theta_w)^s \left\{ \lambda_{t+s} \left(\frac{U_{i,t}}{P_t} \right) \left(\frac{P_t}{P_{t+s}} \right) \left(\frac{U_{i,t}}{W_t} \right)^{-\varepsilon_w} \left(\frac{W_t}{W_{t+s}} \right)^{-\varepsilon_w} H_{t+s} - \frac{\varepsilon_w}{\varepsilon_w - 1} \psi_L \left(\frac{U_{i,t}}{W_t} \right)^{-\varepsilon_w(1+\sigma)} \left(\frac{W_t}{W_{t+s}} \right)^{-\varepsilon_w(1+\sigma)} H_{t+s}^{1+\sigma} \right\} &= 0 \\ \iff \\ \mathbb{E}_t \sum_{s=0}^{\infty} (\beta \theta_w)^s \left\{ \lambda_{t+s} \left(\frac{U_{i,t}}{P_t} \right) \left(\frac{P_t}{P_{t+s}} \right) \left(\frac{W_t}{W_{t+s}} \right)^{-\varepsilon_w} H_{t+s} \right\} \\ = \mathbb{E}_t \sum_{s=0}^{\infty} (\beta \theta_w)^s \left\{ \frac{\varepsilon_w}{\varepsilon_w - 1} \psi_L \left(\frac{U_{i,t}}{W_t} \right)^{-\varepsilon_w \sigma} \left(\frac{W_t}{W_{t+s}} \right)^{-\varepsilon_w(1+\sigma)} H_{t+s}^{1+\sigma} \right\}. \end{aligned}$$

Then,

$$\frac{U_{i,t}}{P_t} = \frac{\varepsilon_w}{\varepsilon_w - 1} \psi_L \left(\frac{U_{i,t}}{W_t} \right)^{-\varepsilon_w \sigma} \frac{\mathbb{E}_t \sum_{s=0}^{\infty} (\beta \theta_w)^s \left\{ \left(\frac{W_t}{W_{t+s}} \right)^{-\varepsilon_w(1+\sigma)} H_{t+s}^{1+\sigma} \right\}}{\mathbb{E}_t \sum_{s=0}^{\infty} (\beta \theta_w)^s \left\{ \lambda_{t+s} \left(\frac{P_t}{P_{t+s}} \right) \left(\frac{W_t}{W_{t+s}} \right)^{-\varepsilon_w} H_{t+s} \right\}}.$$

Note that all unions resetting wages in period t face an identical problem and therefore choose the same reset wage, U_t . Therefore, we can write the optimal real reset wage as:

$$\left(\frac{U_t}{P_t} \right)^{1+\varepsilon_w \sigma} = \frac{\varepsilon_w}{\varepsilon_w - 1} \psi_L \left(\frac{W_t}{P_t} \right)^{\varepsilon_w \sigma} \frac{F_{w,t}}{K_{w,t}}, \quad (\text{A.11})$$

where

$$\begin{aligned} F_{w,t} &= \mathbb{E}_t \sum_{s=0}^{\infty} (\beta \theta_w)^s \left[\left(\frac{W_t}{W_{t+s}} \right)^{-\varepsilon_w(1+\sigma)} H_{t+s}^{1+\sigma} \right] \\ &= H_t^{1+\sigma} + \mathbb{E}_t \sum_{s=1}^{\infty} (\beta \theta_w)^s \left[\left(\frac{W_t}{W_{t+s}} \right)^{-\varepsilon_w(1+\sigma)} H_{t+s}^{1+\sigma} \right] \\ &= H_t^{1+\sigma} + \beta \theta_w \mathbb{E}_t \left(\frac{W_t}{W_{t+1}} \right)^{-\varepsilon_w(1+\sigma)} \sum_{s=0}^{\infty} (\beta \theta_w)^s \left[\left(\frac{W_{t+1}}{W_{t+1+s}} \right)^{-\varepsilon_w(1+\sigma)} H_{t+1+s}^{1+\sigma} \right] \\ &= H_t^{1+\sigma} + \beta \theta_w \mathbb{E}_t \left(\frac{W_t}{W_{t+1}} \right)^{-\varepsilon_w(1+\sigma)} F_{w,t+1} \\ &= H_t^{1+\sigma} + \beta \theta_w \mathbb{E}_t \pi_{w,t+1}^{\varepsilon_w(1+\sigma)} F_{w,t+1} \end{aligned}$$

and

$$\begin{aligned} K_{w,t} &= \mathbb{E}_t \sum_{s=0}^{\infty} (\beta \theta_w)^s \left[\lambda_{t+s} \left(\frac{P_t}{P_{t+s}} \right) \left(\frac{W_t}{W_{t+s}} \right)^{-\varepsilon_w} H_{t+s} \right] \\ &= \lambda_t H_t + \mathbb{E}_t \sum_{s=1}^{\infty} (\beta \theta_w)^s \left[\lambda_{t+s} \left(\frac{P_t}{P_{t+s}} \right) \left(\frac{W_t}{W_{t+s}} \right)^{-\varepsilon_w} H_{t+s} \right] \\ &= \lambda_t H_t + \beta \theta_w \mathbb{E}_t \left(\frac{P_t}{P_{t+1}} \right) \left(\frac{W_t}{W_{t+1}} \right)^{-\varepsilon_w} \sum_{s=0}^{\infty} (\beta \theta_w)^s \left[\lambda_{t+1+s} \left(\frac{P_{t+1}}{P_{t+1+s}} \right) \left(\frac{W_{t+1}}{W_{t+1+s}} \right)^{-\varepsilon_w} H_{t+1+s} \right] \\ &= \lambda_t H_t + \beta \theta_w \mathbb{E}_t \left(\frac{P_t}{P_{t+1}} \right) \left(\frac{W_t}{W_{t+1}} \right)^{-\varepsilon_w} \sum_{s=0}^{\infty} (\beta \theta_w)^s K_{w,t+1} \\ &= \lambda_t H_t + \beta \theta_w \mathbb{E}_t \frac{\pi_{w,t+1}^{\varepsilon_w}}{\pi_{t+1}} K_{w,t+1}. \end{aligned}$$

In case of frictionless labor markets, equation (A.11) simplifies to:

$$\begin{aligned}\left(\frac{U_t}{P_t}\right) &= \frac{\varepsilon_w}{\varepsilon_w - 1} \psi_L \left(\frac{U_t}{W_t}\right)^{-\varepsilon_w \sigma} \frac{H_t^{1+\sigma}}{\lambda_t H_t} \\ &= \frac{\varepsilon_w}{\varepsilon_w - 1} \psi_L \left(\frac{U_t}{W_t}\right)^{-\varepsilon_w \sigma} \frac{h_{i,t}^\sigma}{\lambda_t} \left(\frac{U_t}{W_t}\right)^{-\varepsilon_w \sigma} \\ &= \frac{\varepsilon_w}{\varepsilon_w - 1} \frac{\psi_L h_{i,t}^\sigma}{\lambda_t},\end{aligned}$$

where the second equality exploited the demand curve for labor type i .

Exploiting the fact that all unions resetting wages in period t set the same wage and the Calvo (1983) assumption that the probability of being able to reset the wage rate is independent of time and across labor types, we can write the aggregate wage as a weighted average of the reset wage and last periods aggregate wage where the weights are given by the fraction (which equals the overall probability that a given union is able to reset the wage rate) of unions which adjust their wage in period t :

$$W_t^{1-\varepsilon_w} = (1 - \theta_w) U_t^{1-\varepsilon_w} + \theta_w W_{t-1}^{1-\varepsilon_w}$$

and

$$\left(\frac{W_t}{P_t}\right)^{1-\varepsilon_w} := w_t^{1-\varepsilon_w} = (1 - \theta_w) \left(\frac{U_t}{P_t}\right)^{1-\varepsilon_w} + \theta_w \left(\frac{W_{t-1}}{P_{t-1}}\right)^{1-\varepsilon_w} \left(\frac{P_t}{P_{t-1}}\right)^{\varepsilon_w - 1}.$$

Wage inflation $\pi_{w,t+1}$ can be expressed as:

$$\frac{W_{t+1}}{W_t} := \pi_{w,t+1} = \frac{W_{t+1}/P_{t+1}}{W_t/P_t} \frac{P_{t+1}}{P_t} = \frac{w_{t+1}}{w_t} \pi_{t+1},$$

and

$$\pi_{t+1} := \frac{P_{t+1}}{P_t}$$

is price inflation.

A.5 Optimal Reset Price

Recall:

$$\begin{aligned}C_{kj,t} &= f(k)^{-1} C_{k,t} \left(\frac{P_{kj,t}}{P_{k,t}}\right)^{-\varepsilon_{ck}} \\ C_{k,t} &= f(k) C_t \left(\frac{P_{k,t}}{P_t}\right)^{-\varepsilon_c}.\end{aligned}$$

Therefore, by imposing market clearing in the good markets, we can write:

$$Y_{kj,t} = \left(\frac{P_{kj,t}}{P_{k,t}}\right)^{-\varepsilon_{ck}} \left(\frac{P_{k,t}}{P_t}\right)^{-\varepsilon_c} Y_t.$$

The problem of a firm j in sector k which is able to reoptimize in period t is then to

choose $X_{kj,t}$ to maximize:

$$\begin{aligned} \mathbb{E}_t \sum_{s=0}^{\infty} (\beta \theta_k)^s \frac{\Lambda_{t+s}}{\Lambda_t} & \left(X_{kj,t} Y_{kj,t+s} - W_{t+s} H_{kj,t+s} \right) \\ \text{s.t.} \quad Y_{kj,t+s} &= \left(\frac{X_{kj,t}}{P_{k,t+s}} \right)^{-\varepsilon_{ck}} \left(\frac{P_{k,t+s}}{P_{t+s}} \right)^{-\varepsilon_c} Y_{t+s} \\ Y_{kj,t+s} &= A_{t+s} H_{kj,t+s}, \end{aligned}$$

where $Y_{kj,t}$ is the output of firm j in sector k , Λ is the Lagrange multiplier on the household budget constraint, θ_k is the probability with which a firm in sector k is not able to reset its price, W is the aggregate wage rate of homogeneous labor H , and ε_c and ε_{ck} are the elasticities of substitution in consumption between sectoral subcomposites and within sector varieties.

Substituting the constraints into the objective function:

$$\begin{aligned} \mathbb{E}_t \sum_{s=0}^{\infty} (\beta \theta_k)^s \frac{\Lambda_{t+s}}{\Lambda_t} & \left[X_{kj,t} \left(\frac{X_{kj,t}}{P_{k,t+s}} \right)^{-\varepsilon_{ck}} \left(\frac{P_{k,t+s}}{P_{t+s}} \right)^{-\varepsilon_c} Y_{t+s} \right. \\ & \left. - W_{t+s} \left(\frac{1}{A_{t+s}} \right) \left(\frac{X_{kj,t}}{P_{k,t+s}} \right)^{-\varepsilon_{ck}} \left(\frac{P_{k,t+s}}{P_{t+s}} \right)^{-\varepsilon_c} Y_{t+s} \right] \\ & \iff \\ \mathbb{E}_t \sum_{s=0}^{\infty} (\beta \theta_k)^s \frac{\lambda_{t+s}}{\lambda_t} P_t & \left[\left(\frac{X_{kj,t}}{P_t} \right)^{1-\varepsilon_{ck}} \left(\frac{P_t}{P_{t+s}} \right)^{1-\varepsilon_{ck}} \left(\frac{P_{k,t+s}}{P_{t+s}} \right)^{\varepsilon_{ck}} \left(\frac{P_{k,t+s}}{P_{t+s}} \right)^{-\varepsilon_c} Y_{t+s} \right. \\ & \left. - \left(\frac{W_{t+s}}{P_{t+s}} \right) \left(\frac{1}{A_{t+s}} \right) \left(\frac{X_{kj,t}}{P_t} \right)^{-\varepsilon_{ck}} \left(\frac{P_{k,t+s}}{P_{t+s}} \right)^{\varepsilon_{ck}} \left(\frac{P_{k,t+s}}{P_{t+s}} \right)^{-\varepsilon_c} \left(\frac{P_t}{P_{t+s}} \right)^{-\varepsilon_{ck}} Y_{t+s} \right]. \end{aligned}$$

The first order condition is given by:

$$\begin{aligned} \mathbb{E}_t \sum_{s=0}^{\infty} (\beta \theta_k)^s \frac{\lambda_{t+s}}{\lambda_t} & \left[(1 - \varepsilon_{ck}) \left(\frac{X_{kj,t}}{P_t} \right)^{-\varepsilon_{ck}} \left(\frac{P_{k,t+s}}{P_{t+s}} \right)^{\varepsilon_{ck}} \left(\frac{P_{k,t+s}}{P_{t+s}} \right)^{-\varepsilon_c} \left(\frac{P_t}{P_{t+s}} \right)^{1-\varepsilon_{ck}} Y_{t+s} \right. \\ & \left. + \varepsilon_{ck} \left(\frac{W_{t+s}}{P_{t+s}} \right) \left(\frac{1}{A_{t+s}} \right) \left(\frac{X_{kj,t}}{P_t} \right)^{-\varepsilon_{ck}-1} \left(\frac{P_{k,t+s}}{P_{t+s}} \right)^{\varepsilon_{ck}} \left(\frac{P_{k,t+s}}{P_{t+s}} \right)^{-\varepsilon_c} \left(\frac{P_t}{P_{t+s}} \right)^{-\varepsilon_{ck}} Y_{t+s} \right] \\ & \iff \\ \mathbb{E}_t \sum_{s=0}^{\infty} (\beta \theta_k)^s \frac{\lambda_{t+s}}{\lambda_t} & \left[\left(\frac{X_{kj,t}}{P_t} \right) \left(\frac{P_{k,t+s}}{P_{t+s}} \right)^{\varepsilon_{ck}} \left(\frac{P_{k,t+s}}{P_{t+s}} \right)^{-\varepsilon_c} \left(\frac{P_t}{P_{t+s}} \right)^{1-\varepsilon_{ck}} Y_{t+s} \right] \\ = \mathbb{E}_t \sum_{s=0}^{\infty} (\beta \theta_k)^s \frac{\lambda_{t+s}}{\lambda_t} & \left[\frac{\varepsilon_{ck}}{1 - \varepsilon_{ck}} \left(\frac{W_{t+s}}{P_{t+s}} \right) \left(\frac{1}{A_{t+s}} \right) \left(\frac{P_{k,t+s}}{P_{t+s}} \right)^{\varepsilon_{ck}} \left(\frac{P_{k,t+s}}{P_{t+s}} \right)^{-\varepsilon_c} \left(\frac{P_t}{P_{t+s}} \right)^{-\varepsilon_{ck}} Y_{t+s} \right]. \end{aligned}$$

Rearranging, we get:

$$\frac{X_{kj,t}}{P_t} = \frac{\varepsilon_{ck}}{\varepsilon_{ck} - 1} \frac{\mathbb{E}_t \sum_{s=0}^{\infty} (\beta \theta_k)^s \frac{\lambda_{t+s}}{\lambda_t} \left[\left(\frac{W_{t+s}}{P_{t+s}} \right) \left(\frac{1}{A_{t+s}} \right) \left(\frac{P_{k,t+s}}{P_{t+s}} \right)^{\varepsilon_{ck}} \left(\frac{P_{k,t+s}}{P_{t+s}} \right)^{-\varepsilon_c} \left(\frac{P_t}{P_{t+s}} \right)^{-\varepsilon_{ck}} Y_{t+s} \right]}{\mathbb{E}_t \sum_{s=0}^{\infty} (\beta \theta_k)^s \frac{\lambda_{t+s}}{\lambda_t} \left[\left(\frac{P_{k,t+s}}{P_{t+s}} \right)^{\varepsilon_{ck}} \left(\frac{P_{k,t+s}}{P_{t+s}} \right)^{-\varepsilon_c} \left(\frac{P_t}{P_{t+s}} \right)^{1-\varepsilon_{ck}} Y_{t+s} \right]}.$$

As this expression is independent of firm specific variables, all firms in sector k which are able to reset their prices in period t will choose the identical real price, $\frac{X_{k,t}}{P_t}$:

$$\frac{X_{k,t}}{P_t} := \frac{\varepsilon_{ck}}{1 - \varepsilon_{ck}} \frac{F_{p,k,t}}{K_{p,k,t}},$$

where

$$\begin{aligned} F_{p,k,t} &= \lambda_t \left(\frac{W_t}{P_t} \right) \left(\frac{1}{A_t} \right) \left(\frac{P_{k,t}}{P_t} \right)^{\varepsilon_{ck}} \left(\frac{P_{k,t}}{P_t} \right)^{-\varepsilon_c} Y_t + \beta \theta_k \mathbb{E}_t \left(\frac{P_{t+1}}{P_t} \right)^{\varepsilon_{ck}} F_{p,k,t+1} \\ K_{p,k,t} &= \lambda_t \left(\frac{P_{k,t}}{P_t} \right)^{\varepsilon_{ck}} \left(\frac{P_{k,t}}{P_t} \right)^{-\varepsilon_c} Y_t + \beta \theta_k \mathbb{E}_t \left(\frac{P_{t+1}}{P_t} \right)^{\varepsilon_{ck}-1} K_{p,k,t+1}. \end{aligned}$$

A.6 Aggregate Output

Technically, the model does not possess an aggregate production function in the sense that knowing the state of technology, A_t , and the number of people working, L_t , is not sufficient to determine aggregate output, Y_t . Frictions in the labor market and price frictions lead to distortions in the optimal allocation of resources implying that aggregate output will be generally lower than implied by a frictionless model, $Y_t \leq A_t L_t$. In the following, I derive the “aggregate production function.”

Aggregate labor supply in period t is given by:

$$L_t = \int_0^1 h_{i,t} di = \int_0^1 \left(\frac{W_{i,t}}{W_t} \right)^{-\varepsilon_w} H_t di = H_t D S_{w,t},$$

where wage dispersion $DS_{w,t}$ is defined as:

$$\begin{aligned}
DS_{w,t} &= \int_0^1 \left(\frac{W_{i,t}}{W_t} \right)^{-\varepsilon_w} di \\
&= \int_{wage \text{ adjuster}} \left(\frac{W_{i,t}}{W_t} \right)^{-\varepsilon_w} di + \int_{wage \text{ non-adjuster}} \left(\frac{W_{i,t}}{W_t} \right)^{-\varepsilon_w} di \\
&= \left[(1 - \theta_w) \left(\frac{U_t}{W_t} \right)^{-\varepsilon_w} + \int_{wage \text{ non-adjuster}} \left(\frac{W_{i,t}}{W_t} \right)^{-\varepsilon_w} di \right] \\
&= \left[(1 - \theta_w) \left(\frac{U_t}{W_t} \right)^{-\varepsilon_w} + \theta_w \left(\frac{W_{t-1}}{W_t} \right)^{-\varepsilon_w} \int_0^1 \left(\frac{W_{i,t-1}}{W_{t-1}} \right)^{-\varepsilon_w} di \right] \\
&= \left[(1 - \theta_w) \left(\frac{U_t}{W_t} \right)^{-\varepsilon_w} + \theta_w \left(\frac{W_{t-1}}{W_t} \right)^{-\varepsilon_w} DS_{w,t-1} \right] \\
&= \left[(1 - \theta_w) \left(\frac{U_t}{P_t} \right)^{-\varepsilon_w} \left(\frac{W_t}{P_t} \right)^{\varepsilon_w} + \theta_w \pi_w^{\varepsilon_w} DS_{w,t-1} \right].
\end{aligned}$$

Aggregate labor demand in period t is given by:

$$\begin{aligned}
H_t &= \int_0^1 \int_0^1 H_{kj,t} dj dk \\
&= \int_0^1 \int_0^1 \frac{Y_{kj,t}}{A_t} dj dk \\
&= \frac{1}{A_t} Y_t \int_0^1 \int_0^1 \left(\frac{P_{kj,t}}{P_{k,t}} \right)^{-\varepsilon_{ck}} \left(\frac{P_{k,t}}{P_t} \right)^{-\varepsilon_c} dj dk \\
&= \frac{1}{A_t} Y_t \int_0^1 f(k) DS_{p,k,t} \left(\frac{P_{k,t}}{P_t} \right)^{-\varepsilon_c} dk \\
&= \frac{1}{A_t} Y_t DS_{p,t},
\end{aligned}$$

where price dispersion in sector k $DS_{p,k,t}$ is given by:

$$\begin{aligned}
DS_{p,k,t} &= \int_0^1 \left(\frac{P_{kj,t}}{P_{k,t}} \right)^{-\varepsilon_{ck}} dj \tag{A.12} \\
&= \int_{price \text{ adjuster}} \left(\frac{P_{kj,t}}{P_{k,t}} \right)^{-\varepsilon_{ck}} dj + \int_{price \text{ non-adjuster}} \left(\frac{P_{kj,t}}{P_{k,t}} \right)^{-\varepsilon_{ck}} dj \\
&= \left[(1 - \theta_k) \left(\frac{X_{k,t}}{P_{k,t}} \right)^{-\varepsilon_{ck}} + \theta_k \left(\frac{P_{k,t-1}}{P_{k,t}} \right)^{-\varepsilon_{ck}} DS_{p,k,t-1} \right] \\
&= \left[(1 - \theta_k) \left(\frac{X_{k,t}}{P_t} \right)^{-\varepsilon_{ck}} \left(\frac{P_t}{P_{k,t}} \right)^{-\varepsilon_{ck}} + \theta_k \left(\frac{P_{k,t-1}}{P_{t-1}} \right)^{-\varepsilon_{ck}} \left(\frac{P_{t-1}}{P_t} \right)^{-\varepsilon_{ck}} \left(\frac{P_t}{P_{k,t}} \right)^{-\varepsilon_{ck}} DS_{p,k,t-1} \right]
\end{aligned}$$

and aggregate price dispersion $DS_{p,t}$ by:

$$DS_{p,t} = \int_0^1 f(k) \left(\frac{P_{k,t}}{P_t} \right)^{-\varepsilon_c} DS_{p,k,t} dk. \tag{A.13}$$

Hence, aggregate output in period t is given by:

$$Y_t = \frac{A_t L_t}{DS_{p,t} DS_{w,t}}.$$

Price and wage dispersion, $DS_{p,t}$ and $DS_{w,t}$, will be larger than 1 away from the zero inflation steady state. For instance, for wage dispersion:

$$\begin{aligned} DS_{w,t} &= \int_0^1 \left(\frac{W_{i,t}}{W_t} \right)^{-\varepsilon_w} di \\ &= \int_0^1 \left[\left(\frac{W_{i,t}}{W_t} \right)^{1-\varepsilon_w} \right]^{\frac{-\varepsilon_w}{1-\varepsilon_w}} di \\ &\geq \int_0^1 \left[\left(\frac{W_{i,t}}{W_t} \right)^{1-\varepsilon_w} di \right]^{\frac{-\varepsilon_w}{1-\varepsilon_w}} \\ &= 1^{\frac{-\varepsilon_w}{1-\varepsilon_w}} \\ &= 1, \end{aligned}$$

where the inequality follows from Jensen's inequality and the penultimate equality is due to the definition of the aggregate wage rate:

$$W_t^{1-\varepsilon_w} = \left[\int_0^1 W_{i,t}^{1-\varepsilon_w} di \right].$$

Hence, as stated at the beginning of this section, output will be generally inefficiently low due to distortions in the labor and product markets.

A.7 Price of Claim on Sector Dividends

Real dividends of sector k are given by:

$$D_{k,t} = \frac{1}{P_t} \int_0^1 (P_{kj,t} Y_{kj,t} - H_{kj,t} W_t) dj.$$

Recall that demand of firm j in sector k and the production function are given by:

$$\begin{aligned} Y_{kj,t} &= \left(\frac{P_{kj,t}}{P_{k,t}} \right)^{-\varepsilon_{ck}} \left(\frac{P_{k,t}}{P_t} \right)^{-\varepsilon_c} Y_t \\ H_{kj,t} &= \frac{Y_{kj,t}}{A_t}. \end{aligned}$$

Therefore,

$$\begin{aligned}
D_{k,t} &= \int_0^1 \left[\left(\frac{P_{kj,t}}{P_{k,t}} \right)^{1-\epsilon_{ck}} \left(\frac{P_{k,t}}{P_t} \right)^{1-\epsilon_c} Y_t - \left(\frac{W_t}{P_t} \right) \left(\frac{1}{A_t} \right) \left(\frac{P_{kj,t}}{P_{k,t}} \right)^{-\epsilon_{ck}} \left(\frac{P_{k,t}}{P_t} \right)^{-\epsilon_k} Y_t \right] dj \\
&= \left(\frac{P_{k,t}}{P_t} \right)^{1-\epsilon_c} Y_t \left[\int_0^1 \left(\frac{P_{kj,t}}{P_{k,t}} \right)^{1-\epsilon_{ck}} dj - \left(\frac{W_t}{P_t} \right) \left(\frac{1}{A_t} \right) \left(\frac{P_{k,t}}{P_t} \right)^{-1} \int_0^1 \left(\frac{P_{kj,t}}{P_{k,t}} \right)^{-\epsilon_{ck}} dj \right] \\
&= \left(\frac{P_{k,t}}{P_t} \right)^{1-\epsilon_c} Y_t \left[1 - \left(\frac{W_t}{P_t} \right) \left(\frac{1}{A_t} \right) \left(\frac{P_{k,t}}{P_t} \right)^{-1} DS_{p,k,t} \right] \tag{A.14}
\end{aligned}$$

$$= \left(\frac{P_{k,t}}{P_t} \right)^{1-\epsilon_c} Y_t \left[1 - \frac{1}{\mu_{k,t}} \right], \tag{A.15}$$

where I made use of the definition of the sectoral price index, equation (A.4), the sectoral price dispersion, and equation (A.12); $\mu_{k,t}$ is the markup in sector k . Note that the terms in brackets can be interpreted as sector k profit margin.

Hence, we can write the price of a claim to sector k dividends as follows:

$$\begin{aligned}
P_{k,t} &= \mathbb{E}_t \sum_{s=0}^{\infty} \beta^s \frac{\lambda_{t+s}}{\lambda_t} D_{k,t+s} \\
&= D_{k,t} + \mathbb{E}_t \sum_{s=1}^{\infty} \beta^s \frac{\lambda_{t+s}}{\lambda_t} D_{k,t+s} \\
&= D_{k,t} + \mathbb{E}_t \beta \frac{\lambda_{t+1}}{\lambda_t} \sum_{s=0}^{\infty} \beta^s \frac{\lambda_{t+1+s}}{\lambda_{t+1}} D_{k,t+1+s} \\
&= D_{k,t} + \mathbb{E}_t \beta \frac{\lambda_{t+1}}{\lambda_t} P_{k,t+1}.
\end{aligned}$$

A.8 Cross Sectional Stock Returns at the Sector Level

Starting from the Euler equation or the “central asset pricing formula” (see Cochrane (2005) chapter one) and assuming log-normality, I first derive the expected excess return of a generic asset and then the expected return and return difference across sectors of a claim to sector dividends, which pays off only one period in the future to develop intuition for the key driving forces of the observed cross sectional return premium for sticky-price firms.

$$1 = \mathbb{E}_t[M_{t,t+1}R_{t+1}],$$

where $M_{t,t+1}$ is the stochastic discount factor to price asset between periods t and $t+1$ and R_{t+1} is the gross return of any generic asset.

Exploiting the assumption of log-normality, I get:

$$1 = \exp \left[\mathbb{E}_t m_{t,t+1} + \mathbb{E}_t r_{t+1} + \frac{1}{2} \text{var}_t m_{t,t+1} + \frac{1}{2} \text{var}_t r_{t+1} + \text{cov}_t(m_{t,t+1}, r_{t+1}) \right],$$

where lower case variables correspond to the natural logarithm of upper case variables. Taking logs and subtracting the expression for the risk-free rate, r_{t+1}^f , I can write log expected excess returns as

$$\mathbb{E}_t r_{t+1} - r_{t+1}^f + \frac{1}{2} \text{var } r_{t+1} = -\text{cov}_t(m_{t+1}, r_{t+1}).$$

Consider the return to a claim to sector k dividends in period $t + 1$ with current price $P_{k,t}$, $R_{k,t+1} = D_{k,t+1}/P_{k,t}$.

I can write the expected excess return of a claim to sector one dividends over the return of a claim to sector two dividends (plus Jensen's inequality terms) as:

$$\begin{aligned} \mathbb{E}_t r_{1,t+1} - \mathbb{E}_t r_{2,t+1} + \left[\frac{1}{2} \text{var } r_{1,t+1} - \frac{1}{2} \text{var } r_{2,t+1} \right] &= -\text{cov}_t(m_{t+1}, r_{1,t+1} - r_{2,t+1}) \\ &= -\text{cov}_t(m_{t+1}, d_{1,t+1} - d_{2,t+1}). \end{aligned}$$

Using equation (A.15), I can write the log difference in dividends as:

$$d_{1,t+1} - d_{2,t+1} = (1 - \varepsilon_c)(p_{1,t+1} - p_{2,t+1}) + \left[\log \left(1 - \frac{1}{\mu_{1,t+1}} \right) - \log \left(1 - \frac{1}{\mu_{2,t+1}} \right) \right].$$

Taking a first-order Taylor series approximation around the steady state markup, $\mu_1 = \mu_2 = \varepsilon_{ck}/(\varepsilon_{ck} - 1)$:

$$\log \left(1 - \frac{1}{\mu_{k,t+1}} \right) \approx \log \left(1 - \frac{1}{\mu_k} \right) + \frac{1}{\mu_k - 1} (\log \mu_{k,t+1} - \log \mu_k).$$

Hence, I can write differences in log dividends as:

$$d_{1,t+1} - d_{2,t+1} = (1 - \varepsilon_c)(p_{1,t+1} - p_{2,t+1}) + (\varepsilon_{ck} - 1)(\log \mu_{1,t+1} - \log \mu_{2,t+1}).$$

Making use of equation (A.14), I can express this relation as:

$$d_{1,t+1} - d_{2,t+1} = (1 - \varepsilon_c)(p_{1,t+1} - p_{2,t+1}) + (\varepsilon_{ck} - 1)[(p_{1,t+1} - p_{2,t+1}) - (ds_{p,1,t+1} - ds_{p,2,t+1})].$$

Thus, expected excess returns are given by:

$$\begin{aligned} \mathbb{E}_t r_{1,t+1} - \mathbb{E}_t r_{2,t+1} + \left[\frac{1}{2} \text{var } r_{1,t+1} - \frac{1}{2} \text{var } r_{2,t+1} \right] &= -(\varepsilon_{ck} - \varepsilon_c) \text{cov}_t(m_{t,t+1}, (p_{1,t+1} - p_{2,t+1})) \\ &\quad - (1 - \varepsilon_{ck}) \text{cov}_t(m_{t,t+1}, (ds_{p,1,t+1} - ds_{p,2,t+1})). \end{aligned}$$

B. Summary of Equilibrium Conditions

B.1 Sector Specific Conditions

Reset Price

$$\frac{X_{k,t}}{P_t} = \frac{\varepsilon_{ck}}{\varepsilon_{ck} - 1} \frac{F_{p,k,t}}{K_{p,k,t}}$$

$$F_{p,k,t} = \lambda_t \left(\frac{W_t}{P_t} \right) \left(\frac{1}{A_t} \right) \left(\frac{P_{k,t}}{P_t} \right)^{\varepsilon_{ck}} \left(\frac{P_{k,t}}{P_t} \right)^{-\varepsilon_c} Y_t + \beta \theta_k \left(\frac{P_{t+1}}{P_t} \right)^{\varepsilon_{ck}} F_{p,k,t+1}$$

$$K_{p,k,t} = \lambda_t \left(\frac{P_{k,t}}{P_t} \right)^{\varepsilon_{ck}} \left(\frac{P_{k,t}}{P_t} \right)^{-\varepsilon_c} Y_t + \beta \theta_k \left(\frac{P_{t+1}}{P_t} \right)^{\varepsilon_{ck}-1} K_{p,k,t+1}$$

Relative Price

$$P_{k,t} = \left[(1 - \theta_k) X_{k,t}^{1-\varepsilon_{ck}} + \theta_k P_{k,t-1}^{1-\varepsilon_{ck}} \right]^{\frac{1}{1-\varepsilon_{ck}}}$$

Price Dispersion

$$DS_{p,k,t} = \left[(1 - \theta_k) \left(\frac{X_{k,t}}{P_t} \right)^{-\varepsilon_{ck}} \left(\frac{P_t}{P_{k,t}} \right)^{-\varepsilon_{ck}} + \theta_k \left(\frac{P_{k,t-1}}{P_{t-1}} \right)^{-\varepsilon_{ck}} \left(\frac{P_{t-1}}{P_t} \right)^{-\varepsilon_{ck}} \left(\frac{P_t}{P_{k,t}} \right)^{-\varepsilon_{ck}} DS_{p,k,t-1} \right]$$

B.2 Aggregate Conditions

Law of Motion for Price Level

$$1 = \int_0^1 f(k) \left[(1 - \theta_k) \left(\frac{X_{k,t}}{P_t} \right)^{(1-\varepsilon_c)} + \theta_k \left(\frac{P_{k,t-1}}{P_{t-1}} \right)^{(1-\varepsilon_c)} \left(\frac{P_{t-1}}{P_t} \right)^{(1-\varepsilon_c)} \right] dk$$

Aggregate Price dispersion

$$DS_{p,t} = \int_0^1 f(k) \left(\frac{P_{k,t}}{P_t} \right)^{-\varepsilon_c} DS_{p,k,t} dk.$$

Reset Wage

$$\begin{aligned} \left(\frac{U_t}{P_t} \right)^{1+\varepsilon_w \sigma} &= \frac{\varepsilon_w}{\varepsilon_w - 1} \psi_L \left(\frac{W_t}{P_t} \right)^{\varepsilon_w \sigma} \frac{F_{w,t}}{K_{w,t}} \\ F_{w,t} &= H_t^{1+\sigma} + \beta \theta_w \mathbb{E}_t \pi_{w,t+1}^{\varepsilon_w(1+\sigma)} F_{w,t+1} \\ K_{w,t} &= \lambda_t H_t + \beta \theta_w \mathbb{E}_t \frac{\pi_{w,t+1}^{\varepsilon_w}}{\pi_{t+1}} K_{w,t+1}. \end{aligned}$$

Wage Dispersion

$$DS_{w,t} = \left[(1 - \theta_w) \left(\frac{U_t}{P_t} \right)^{-\varepsilon_w} \left(\frac{W_t}{P_t} \right)^{\varepsilon_w} + \theta_w \pi_w^{\varepsilon_w} DS_{w,t-1} \right]$$

Aggregate Real Wage

$$\frac{W_t}{P_t} = \left[(1 - \theta_w) \left(\frac{U_t}{P_t} \right)^{1-\varepsilon_w} + \theta_w \left(\frac{W_{t-1}}{P_{t-1}} \right)^{1-\varepsilon_w} \left(\frac{P_{t-1}}{P_t} \right)^{1-\varepsilon_w} \right]^{\frac{1}{1-\varepsilon_w}}$$

Wage Inflation

$$\frac{W_{t+1}}{W_t} = \pi_{w,t+1} = \frac{W_{t+1}/P_{t+1}}{W_t/P_t} \frac{P_{t+1}}{P_t} = \frac{w_{t+1}}{w_t} \pi_{t+1}$$

Consumption Euler Equation

$$1 = \beta R_t \mathbb{E}_t \left[\frac{1}{\pi_{t+1}} \left(\frac{C_{t+1} - bC_t}{C_t - bC_{t-1}} \right)^{-\gamma} \right]$$

Aggregate Output

$$Y_t = \frac{A_t L_t}{DS_{p,t} DS_{w,t}}$$

Monetary Policy

$$i_t = \phi_\pi \pi_t + \phi_x x_t + \log \left(\frac{1}{\beta} \right) + u_{m,t}$$

$$u_{m,t} = \rho_m u_{m,t-1} + \sigma_{mp} \varepsilon_{mp,t+1}$$

$$a_{t+1} = \rho_a a_t + \sigma_a \varepsilon_{a,t+1},$$

C. Aggregation of good-based Frequencies of Price Adjustment

In this section I discuss in more detail how I aggregate good-based frequencies of price adjustment to the firm level in a two stage procedure.

I first aggregate goods' based frequencies to the establishment level via internal identifiers of the BLS. To perform the firm level aggregation, I check whether establishments with the same or similar names are part of the same company. In addition, I use publicly available data to search for names of subsidiaries and name changes e.g. due to mergers, acquisitions or restructuring occurring during the sample period for all firms in the dataset.

I discuss the fictitious case of the company Milkwell Inc, to illustrate aggregation to the firm level. Assume I observe product prices of items for the establishments Milkwell Advanced Circuit, Milkwell Aerospace, Milkwell Automation and Control, Milkwell Mint and Bier Good. In the first step, I calculate the frequency of product price adjustment at the item level and aggregate this measure at the establishment level for all of the above mentioned establishments. I calculate both equally weighted frequencies, U and frequencies weighted by values of shipments associated with items/establishments, W , say for establishment Milkwell Aerospace. I then use publicly available information to check whether the individual establishments are part of the same company. Let's assume that I find that all of the above mentioned establishments with Milkwell in the establishment name but Milkwell Mint are part of Milkwell Inc. Looking at the company structure, I also find that Milkwell has several subsidiaries, Honeymoon, Pears and Bier Good. Using this information, I then aggregate the establishment level frequencies of Milkwell Advanced Circuit, Milkwell Aerospace, Milkwell Automation and Control and Bier Good to the company level, again calculating equally weighted and value of shipments weighted frequencies.

D. Definition of Financial Data

In this section I discuss in more detail the construction of financial variables.

Stock return, shares outstanding, and volume data are from the CRSP Monthly Stock file. I focus on firms that have been part of the S&P500 between 1994 and 2009 because of the availability of the PPI data and to keep the manual merging between the

two datasets manageable. Size of year t is the natural logarithm of the total market capitalization at the firm level as of December $t-1$. β (Beta) is the regression coefficient in rolling time-series regressions of monthly excess returns on a constant and the excess returns of the CRSP value-weighted index over a sixty-month period. Turnover is the ratio of volume to shares outstanding (in percent). Spread is the monthly average of the daily bid-ask spreads from the CRSP Daily Stock file.

I obtain balance-sheet data from the Standard and Poor's Compustat database. I define book equity (BE) as total stockholders' equity plus deferred taxes and investment tax credit (if available) minus the book value of preferred stock. Based on availability, I use the redemption value, liquidation value, or par value (in that order) for the book value of preferred stock. I prefer the shareholders' equity number as reported by Compustat. If not available, I calculate shareholders' equity as the sum of common and preferred equity. If neither of the two is available, I define shareholders' equity as the difference between total assets and total liabilities. The book-to-market (BM) ratio of year t is then the book equity for the fiscal year ending in calendar year $t-1$ over the market equity as of December $t-1$. Leverage (Lev) is the ratio of total long-term debt and debt in current liabilities over the sum of the numerator and shareholders' equity. Cash flow (CF) is the sum of income before extraordinary items and depreciation and amortization over total assets. I calculate the price-to-cost margin (PCM) as net sales minus cost of goods sold over net sales and HHI as the Herfindahl-Hirschman index of sales at the Fama & French 48 industry level at an annual frequency.

E. Cash Flow and Discount Rate Betas

In this section I derive the key equations of the Campbell and Vuolteenaho (2004) decomposition of CAPM β into cash flow and discount rate β .

Following Campbell and Shiller (1988) and Campbell (1991), I use a loglinear approximation for the one period log return and iterate the resulting relationship forward to obtain an accounting identity expressing unexpected returns, $r_{t+1} - \mathbb{E}_t r_{t+1}$, as a function of revisions in future dividend growth, Δd_{t+1+s} , and returns

$$\begin{aligned} r_{t+1} - \mathbb{E}_t r_{t+1} &= (\mathbb{E}_{t+1} - \mathbb{E}_t) \sum_{s=0}^{\infty} \rho^s \Delta d_{t+1+s} - (\mathbb{E}_{t+1} - \mathbb{E}_t) \sum_{s=1}^{\infty} \rho^s r_{t+1+s} \\ &= N_{CF,t+1} - N_{DR,t+1}, \end{aligned}$$

where ρ is a discount coefficient slightly below one, $N_{CF,t+1}$ denotes news about future

dividends and $N_{DR,t+1}$ news about future expected returns.¹

To obtain the news terms, I estimate a first-order VAR

$$z_{t+1} = a + \Gamma z_t + u_{t+1}$$

where z_{t+1} is a $m \times 1$ state vector with r_{t+1} as its first element, a and Γ are a $m \times 1$ vector and a $m \times m$ matrix of coefficients and u_{t+1} is a $m \times 1$ vector of one-step ahead forecast errors.²

Cash flow and discount rate news are then functions of $t + 1$ shocks

$$N_{CF,t+1} = (e1' + e1'\lambda)u_{t+1} \quad (\text{A.16})$$

$$N_{DR,t+1} = e1'\lambda u_{t+1}, \quad (\text{A.17})$$

where $e1$ is a $m \times 1$ selection vector whose first element is one and whose other elements are zero, λ is defined as $\lambda \equiv \rho\Gamma(I - \rho\Gamma)^{-1}$ and I is a $m \times m$ identity matrix.

I follow Campbell and Vuolteenaho (2004) and estimate a VAR(1) over the full sample period at a monthly frequency using four state variables: the log excess return on the CRSP value weighted index, the yield spread between the 10-year constant maturity bonds (Global Financial Data (GFD) symbol IGUSA10D) and the 1-year constant maturity notes (GFD symbol IGUSA1D), the 10-year smoothed price-earnings ratio of Shiller (2000) as well as the small stock value spread calculated as in Campbell and Vuolteenaho (2004).³

I define cash flow and discount rate betas as

$$\beta_{p,CF} \equiv \frac{Cov(r_{p,t}^e, N_{CF,t})}{Var(r_{m,t}^e - \mathbb{E}_{t-1} r_{m,t}^e)}$$

$$\beta_{p,DR} \equiv \frac{Cov(r_{p,t}^e, -N_{DR,t})}{Var(r_{m,t}^e - \mathbb{E}_{t-1} r_{m,t}^e)},$$

where $r_{p,t}^e$ is the log excess return of portfolio p and $r_{m,t}^e$ is the log excess return of the market. $\beta_{p,CF}$ and $\beta_{p,DR}$ add up to the CAPM beta, β_p . I estimate these betas using the VAR-fitted news series to construct sample variances and covariances allowing for one additional lag of the news terms. I calculate GMM standard errors conditional on the realized news series from the VAR.⁴

¹See Campbell and Vuolteenaho (2004) for an interpretation of ρ . I follow them and set ρ to a value of 0.95 at an annual frequency.

²This formulation is not restrictive as any higher-order VAR can be written in its companion form as a VAR(1) by suitably augmenting the state vector.

³This is the VAR specification as in the original contribution of Campbell and Vuolteenaho (2004).

⁴Specifically, I write the covariances and variances as functions of sample means and apply the Delta method as advocated in chapter 11 of Cochrane (2005).

F. Additional Results

This section contains several robustness checks and results for alternative measures of the frequency of price adjustment.

As discussed in the main body of Weber (2015), I calculate the frequency of price adjustment as the mean fraction of months with price changes during the sample period of an item. Because the collected data may have missing values, I construct different measures of the frequency of price adjustment, S . In the first approach, labeled A , I treat missing values as interrupting price spells. For example, if a price was \$4 for two months, then misses for a month, and is again observed at \$5 for another three months, I treat the data as reporting two price spells with durations of two and three months where none of the spells has a price change and hence the frequency is zero. In the second approach, labeled B , missing values do not interrupt price histories. In the previous example, approach B concatenates spells of \$4 and \$5 prices and yields one price change in five months so that the frequency is $1/5$. Approach C takes the union of A and B ; that is, there is a price change if either A or B identify a price change. I employ approach SA in the main paper weighting item-based frequencies equally (U). Results are very similar if I make use of these alternative measures or weight individual item-based frequencies with their value of shipment (W).

Table A.1 reports mean frequencies, standard deviations, and the number of firm-months observations for these different measures of the frequency of price adjustment, both for the total sample and for different industries separately.

Table A.2 reports descriptive statistics for the firm characteristics and return predictors for the benchmark sample period from July 1982 to June 2007 for all firms independent of missing data on the frequency of price adjustment. The full and merged sample (Table 2 of Weber (2015)) is virtually identical based on observables, which is expected given the probabilistic sampling underlying the construction of the producer price index.

Table A.3 runs panel regressions with my measure of price stickiness as left-hand-side variable. Individually, we see that book-to-market, beta, leverage, and the price-to-cost margin are most strongly related with price stickiness. Firms with more flexible prices have lower CAPM beta and price-to-cost margins but higher book-to-market ratios and leverage. Jointly, these firm characteristics can explain up to 12% of the variation in SAU across firms.

Table A.4 contains times series means of portfolio average firm characteristics. Column 1 shows that the frequency of price adjustment is by construction monotonically increasing from as low as 0.01 per month for portfolio 1 to 0.35 for the flexible price portfolio. The following columns document similar facts as the correlations in Table 2: firms with flexible prices are on average larger and have a slightly higher book-to-market ratio, but lower systematic risk as measured by beta. There is no difference in cash flows, turnover, or bid-ask spreads. The last two columns document that part of the difference in the frequency of price adjustment could reflect market power, as the portfolio with flexible price firms has a lower price-to-cost margin and Herfindahl - Hirschman index than the portfolio containing sticky-price firms.

Table A.5 repeats the baseline regression of annual stock returns on firm characteristics and return predictors only. The individual coefficients generally have the expected signs, and most of them are individually statistically significant. R^2 s are small in all firm-level panel regressions and typically below 1%.

I construct a zero-cost portfolio, which invests in stocks with low frequency of price adjustment, and funds this investment by selling short flexible price firms (L-H in the following). Table A.6 reports α s in percent per month and β s for the unconditional CAPM. I evaluate statistical significance using the time-series variability of the slope and intercept coefficients following Fama and MacBeth (1973) in parentheses and Newey and West (1987)-corrected standard errors in brackets. The unconditional CAPM cannot explain the portfolio returns. Monthly α s range between 0.47% and 0.56% per month and are highly statistically significant. In column (6), we also see that the L-H portfolio has a statistically insignificant α of 0.10% per month.

Table A.7 reports results for long-horizon forecasting regressions of log excess returns of the zero-cost portfolio of going long the portfolio of stocks with high book-to-market value of equity and shorting the portfolio of stocks with low book-to-market value of equity of Fama and French (1993), HML, in Panel A, and the CRSP value-weighted excess return in Panel B on the proxy for the consumption-wealth ratio of Lettau and Ludvigson (2001). cay has a marginally statistically significant forecasting power, the log excess return of the CRSP value-weighted index. The maximal R^2 is 35% at a four-year horizon. cay negatively predicts the HML factor of Fama and French (1993) with a maximum explanatory power of 22% at a five-year forecasting horizon; again barely statistically significant however.

Table A.8 report the parameter values used in the model calibration.

Habit formation in consumption implies that expected returns vary over time and are particularly high during recessions. To test this hypothesis, I define recessions and expansions as months in the bottom and top 25th percentile of the GDP growth distribution, respectively, and measure the subsequently realized return spread between sticky- and flexible-price sectors in simulated data. The spread in annual returns in the two years after recessions is 4.1%, whereas it is only 1.1% after expansions, indicating substantial variation in expected returns.

To test more systematically for time variation in expected returns, I run long-horizon regressions on simulated data. I regress the cumulative log excess returns of the L-H portfolio on log consumption surplus. Table A.9 shows the classical patterns: high consumption compared to habit predicts low future excess returns. The regression coefficients increase in absolute value from -0.14 for one-quarter-ahead excess returns to -0.81 for the three-year horizon and then start to decline. The explanatory power peaks at a two-year horizon with consumption-surplus explaining 22% of the time-series variation.

Table A.10 repeats the baseline exercise for the benchmark period for overlapping annual returns at the monthly frequency. Results are quantitatively as well as statistically very similar to my benchmark analysis. None of these variations has a material impact on the coefficient on the frequency of price adjustment; the return premium for sticky price firms varies between 2.8% per annum when including the full set of controls (column (11)) and 7.5% in the specification only controlling for book-to-market (column (2)).

Table A.11 adds the durability of output from Bils et al. (2012) as additional covariate. Controlling for the variation in durability actually slightly increases the premium for sticky-price firms.

In Table A.12, I add a specification in which I cluster standard errors along the time and firm dimensions. While the statistical significance of the coefficient on the frequency of price adjustment only slightly decreases, some other standard return predictors lose their explanatory power. Table A.13 repeats my within-industry results, but adds finer defined industry dummies. In particular, I add dummies at the Fama & French 10 and 17 industry level. As expected, the more I restrict the variation in the data by adding additional fixed effects, the smaller the coefficient on the frequency of price adjustment becomes.

Table A.14 repeats the baseline panel specifications on unwinsorized variables. The coefficient on SAU tends to be larger in absolute value at continuously high statistical significance, while the explanatory power of the model somewhat decreases.

Table A.16 reports results at the portfolio level for a subset of firms which were part of the S&P500 at the end of June 1994 in Panel A, while Panel B chooses 2006 as the cutoff year. Results are very similar to my baseline finding and show that overall premium for sticky price firms is not due to attrition or selection.

Table A.15 reports results at the portfolio level for alternative measures of the frequency of price adjustment, both for raw returns. Return premia for the L-H portfolio range between 2.87% (*SCW*, 1982-2014) and 3.78% (*SAW*, 1982-2007) per annum, similar in magnitude to the numbers reported in the main part of the paper (see Table 3).

The higher riskiness of firms with lower frequencies of price adjustment might potentially also be reflected in higher realized volatilities at the firm level. I construct this measure by taking the square root of the sum of daily squared returns over one year. Table A.17 documents a similar pattern for realized volatilities as we have seen for returns: across specifications – controlling for firm characteristics both individually and jointly – price flexibility is associated with lower realized volatilities, which is highly statistically significant for most estimations. Going from firms with most flexible product prices in sample to firms with stickiest prices is associated with an increase in realized volatility of 6.05% per annum in column (1) for the regression without additional controls and varies between 3.88% in column (12) when I only control for all covariates and 6.49% in column (9) when the bid-ask spread is the sole additional control. The specification controlling for β in column (5) is an exception, as β drives out the measure of price stickiness. This might indicate that β to some extent already captures the higher riskiness of sticky-price firms.⁵ Once I add all controls, though, the coefficient on SAU is again highly statistically significant.

Table A.18 reports descriptive statistics for the firm characteristics and return predictors for the full sample period from July 1982 to June 2014 and Table A.19 the corresponding results of panel regressions of annual stock returns on the measure for the frequency of price adjustment, *SAU*, and controls according to equation (1). Results are very similar to the benchmark sample period, and the regression coefficients imply a premium in returns moving from firms with low frequencies of price adjustment to

⁵This result is somewhat expected as I look at total realized volatilities and not at idiosyncratic realized volatilities controlling for exposure to market risk.

firms with high frequencies of price adjustment between 2.7% in column (12) and 6.6% in column (4).⁶

Figures A.1 and A.2 show why there is only a small risk premium associated with technology shocks. The mean reversion in technology has it that aggregate output moves only little in reaction to the shock, translating into a small reaction in marginal utility and finally dividends, stock prices and returns.

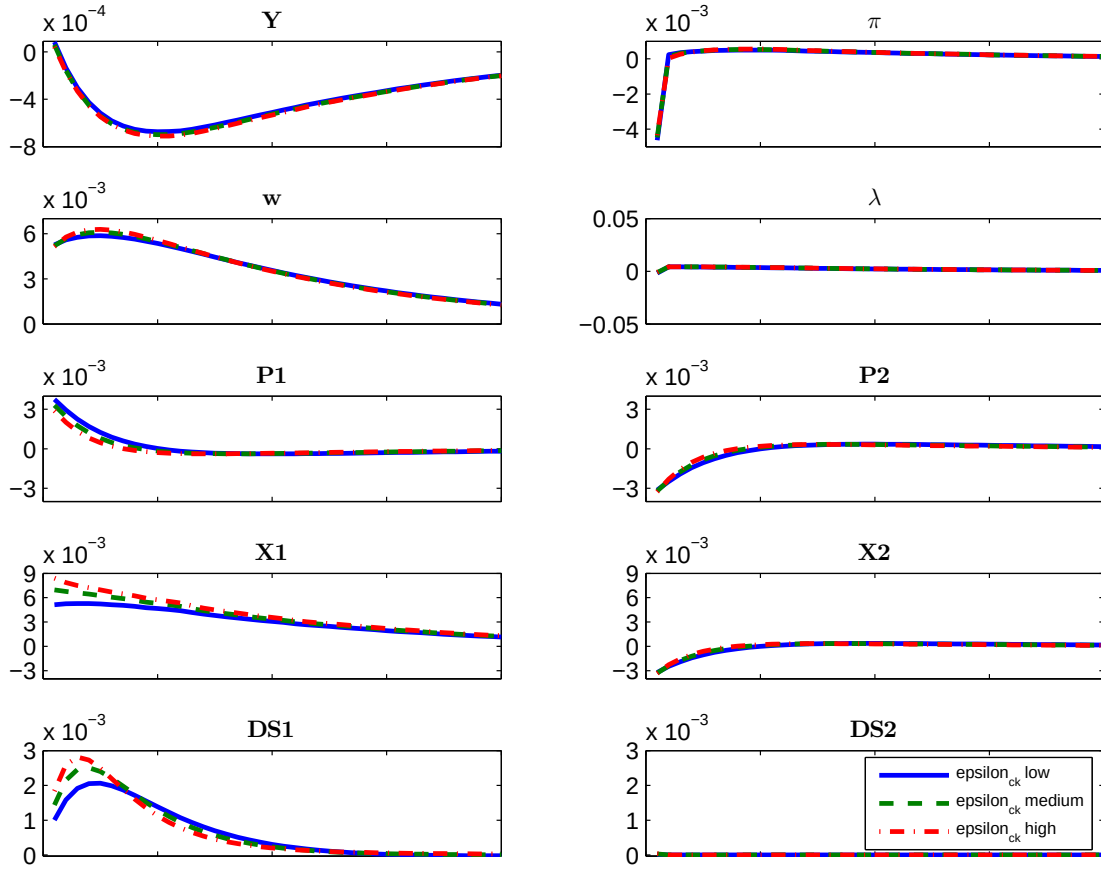
Figures A.3 and A.6 show why variations in the within-sector elasticity of substitutions are quantitatively more important than variations in the elasticity of substitution between sectoral consumption aggregates.⁷ Sector price indices are independent of ε_c and therefore the variation in relative sector price indices we see in Figure A.3 is driven by the effect of ε_c on the aggregate price level when aggregating sector prices. Once we look at differences across sectors, this effect cancels out, and hence variation in ε_c has no effect on the price margin in equation (3). Equation (A.12) shows that sector price dispersion is independent of the across-sector elasticity of substitution in consumption and, therefore, the inefficiency margin is also not impacted by changes in ε_c . Hence, ε_c only affects the quantity margin ($\check{Y}_{1,t} - \check{Y}_{2,t}$). Real sector output is determined by aggregate output times the relative sector price $P_{k,t}/P_t$ to the power of $(1 - \varepsilon_c)$. Increasing the across-sector elasticity of substitution in consumption translates into more negative differences in dividends between the sticky- and flexible-price sector, and therefore increases the cross sectional return difference. This channel, however, is quantitatively small and of second order compared the effects of the within-sector elasticity of substitution.⁸

⁶These premia are calculated by multiplying the regression coefficient on the frequency of price adjustment by 0.6, the differential in the frequency of price adjustment between firms which most infrequently change their product prices and firms which are most flexible in adjusting prices (see Table A.1).

⁷ ε_c low, medium, and high correspond to values of between sectoral consumption aggregates of 4, 8 and 12, respectively. The return of the zero cost portfolio of going long the portfolio of stocks with low frequencies of price adjustment and shorting the flexible price portfolio increases from 2.34% per year to 3.69%.

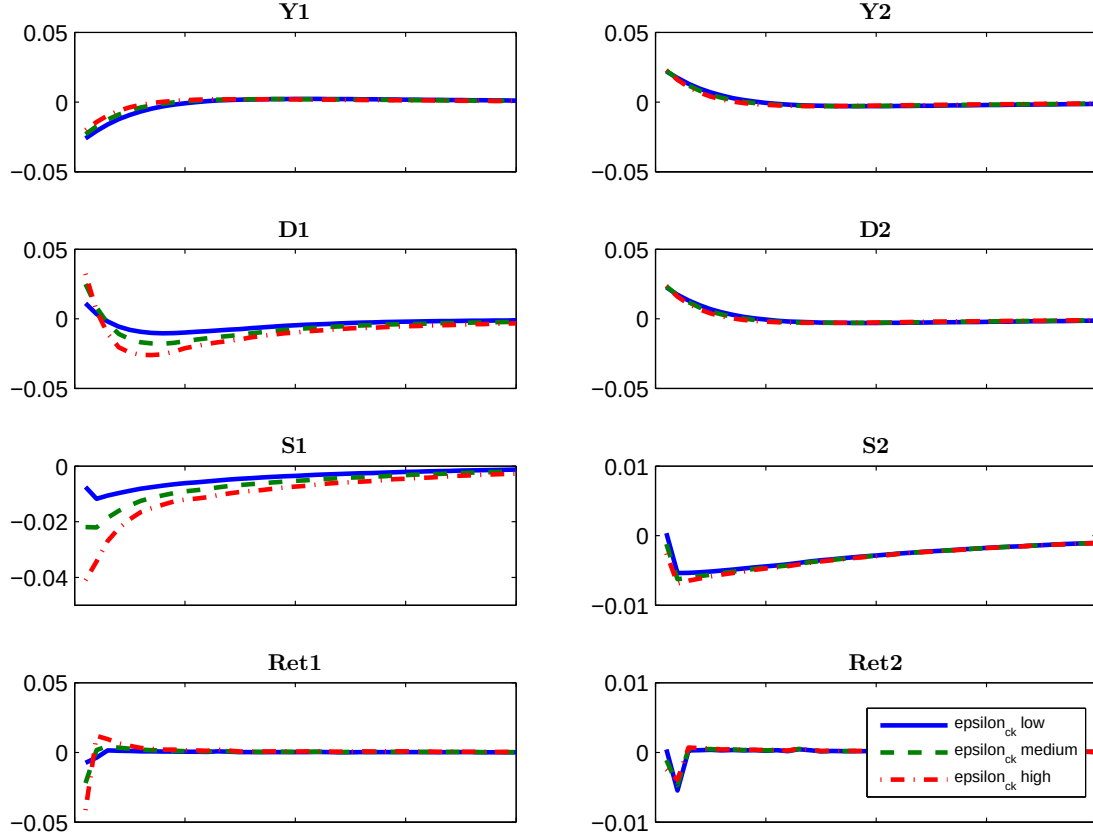
⁸As Figure A.3 shows, changes in ε_c also barely affect aggregate quantities and therefore also have no indirect effects on stock returns via a valuation channel.

Figure A.1: Impulse Response Functions to Technology Shock (varying ε_{ck})



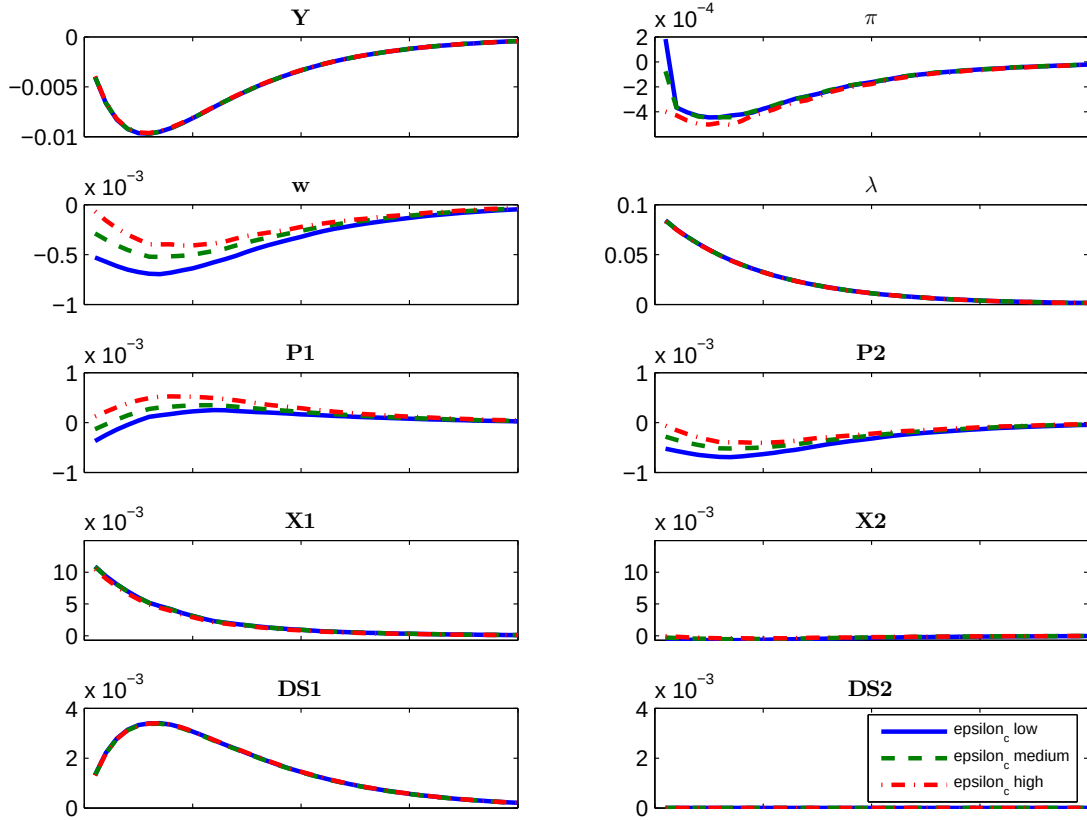
This figure plots the impulse response functions of several macroeconomic variables of a two sector version of the model of Section IV to a one standard deviation technology shock for different values of the elasticity of substitution of within sector consumption varieties, ε_{ck} . ε_{ck} low, medium and high correspond to values of 8, 12 and 16, respectively. Y is output, π inflation, w aggregate real wage, λ the marginal utility of consumption, $P1$ and $P2$ the relative prices of sectors one and two, $X1$ and $X2$ the optimal real reset prices, and $DS1$ and $DS2$ the price dispersion in the two sectors.

Figure A.2: Impulse Response Functions to Technology Shock (varying ε_{ck})



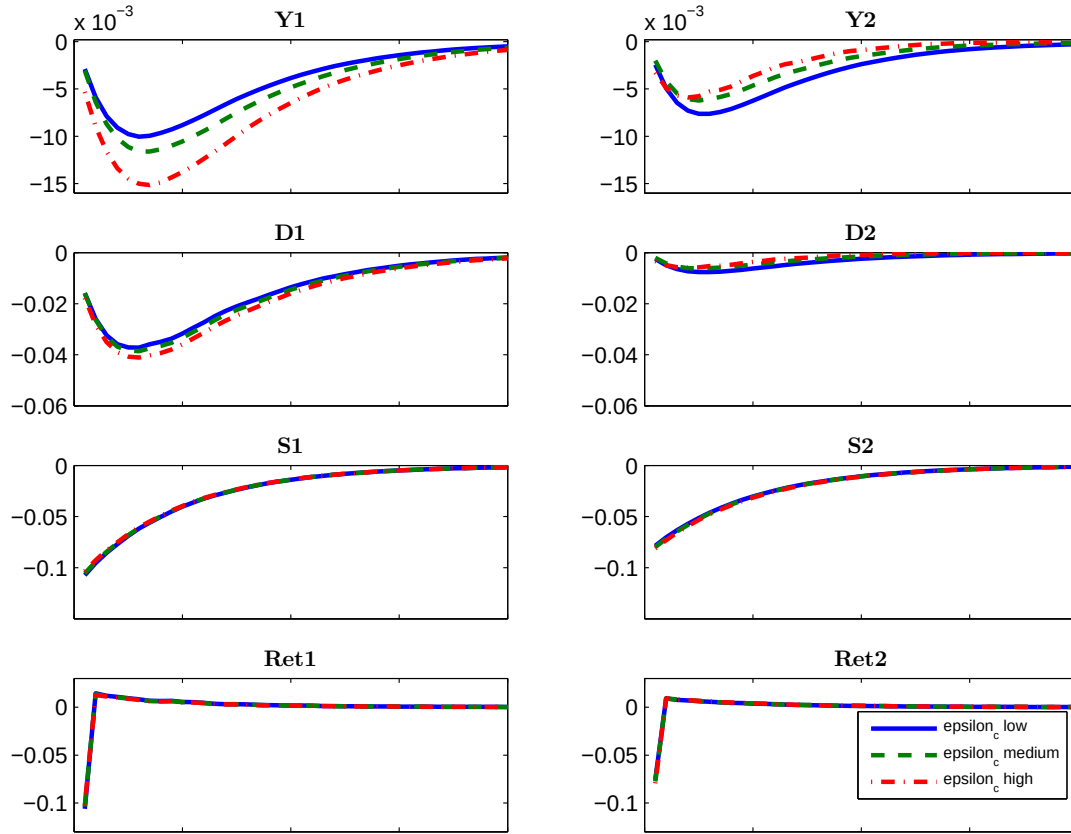
This figure plots the impulse response functions of several macroeconomic variables and asset returns of a two sector version of the model of Section IV to a one standard deviation technology shock for different values of the elasticity of substitution of within sector consumption varieties, ε_{ck} . ε_{ck} low, medium and high correspond to values of 8, 12 and 16, respectively. $Y1$ and $Y2$ are the output of sectors one and two, $D1$ and $D2$ sector level dividends, $S1$ and $S2$ the prices of claims to aggregate sector dividends and $Ret1$ and $Ret2$ the returns of these claims.

Figure A.3: Impulse Response Functions to Monetary Policy Shock (varying ε_c)



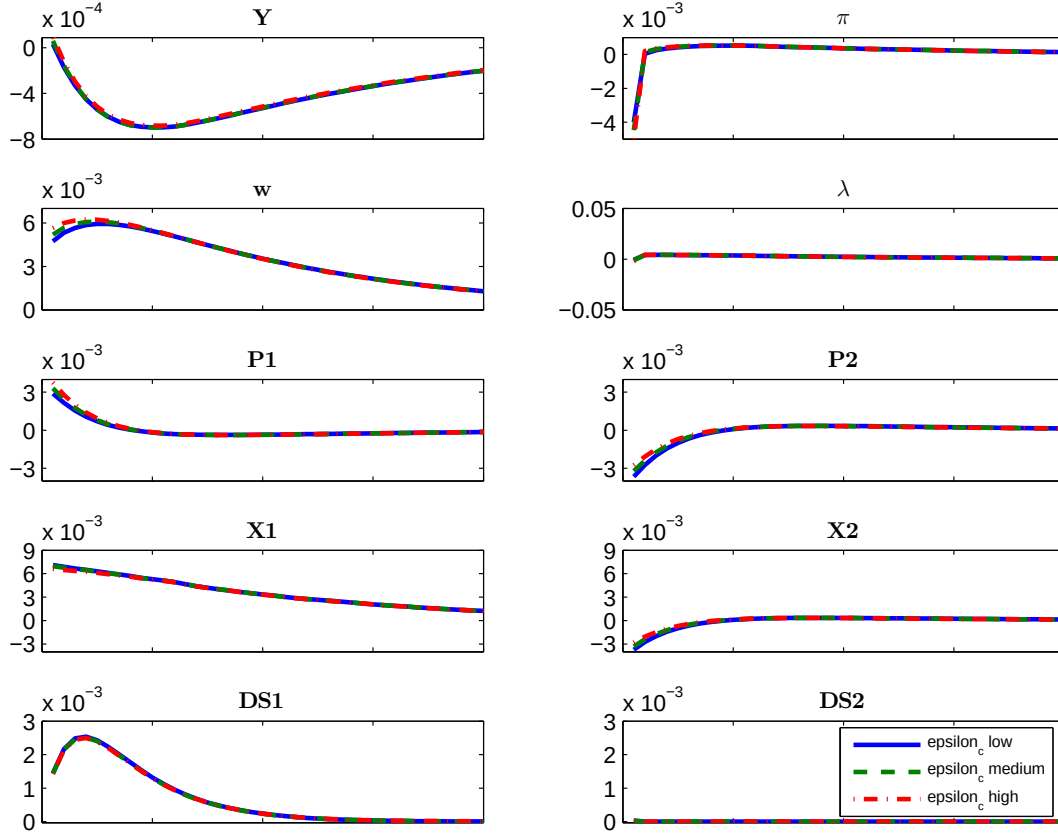
This figure plots the impulse response functions of several macroeconomic variables of the model of Section IV to a one standard deviation monetary policy shock for different values of the elasticity of substitution of across sector consumption varieties, ε_c . ε_c low, medium and high correspond to values of 4, 8 and 12, respectively. Y is output, π inflation, w aggregate real wage, λ the marginal utility of consumption, $P1$ and $P2$ the relative prices of sectors one and two, $X1$ and $X2$ the optimal real reset prices, and $DS1$ and $DS2$ the price dispersion in the two sectors.

Figure A.4: Impulse Response Functions to Monetary Policy Shock (varying ε_c)



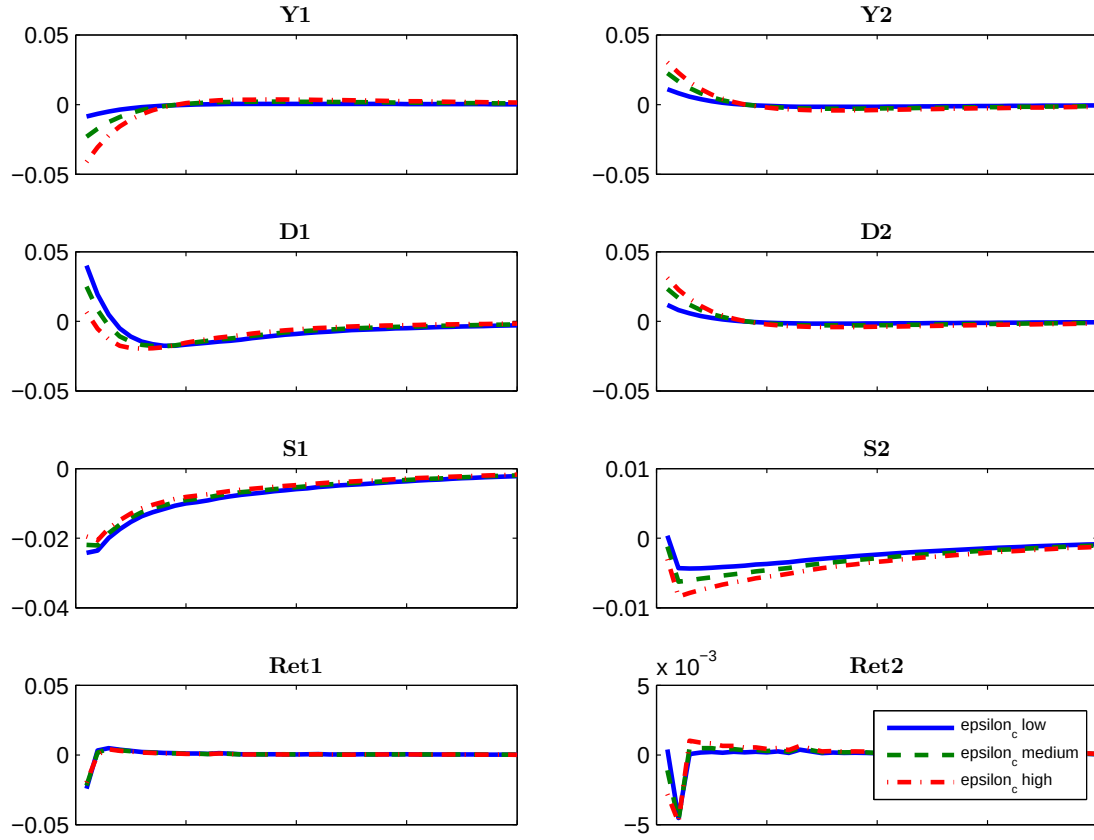
This figure plots the impulse response functions of several macroeconomic variables of the model of Section IV to a one standard deviation monetary policy shock for different values of the elasticity of substitution of across sector consumption varieties, ε_c . ε_c low, medium and high correspond to values of 4, 8 and 12, respectively. Y1 and Y2 are the output of sector one and two, D1 and D2 sector level dividends, S1 and S2 the prices of claims to aggregate sector dividends and Ret1 and Ret2 the returns of these claims.

Figure A.5: Impulse Response Functions to Technology Shock (varying ε_c)



This figure plots the impulse response functions of several macroeconomic variables of the model of Section IV to a one standard deviation technology shock for different values of the elasticity of substitution of across sector consumption varieties, ε_c . ε_c low, medium and high correspond to values of 4, 8 and 12, respectively. Y is output, π inflation, w aggregate real wage, λ the marginal utility of consumption, $P1$ and $P2$ the relative prices of sectors one and two, $X1$ and $X2$ the optimal real reset prices, and $DS1$ and $DS2$ the price dispersion in the two sectors.

Figure A.6: Impulse Response Functions to Technology Shock (varying ε_c)



This figure plots the impulse response functions of several macroeconomic variables of the model of Section IV to a one standard deviation technology shocks for different values of the elasticity of substitution of across sector consumption varieties, ε_c . ε_c low, medium and high correspond to values of 4, 8 and 12, respectively. $Y1$ and $Y2$ are the output of sectors one and two, $D1$ and $D2$ sector level dividends, $S1$ and $S2$ the prices of claims to aggregate sector dividends and $Ret1$ and $Ret2$ the returns of these claims.

Table A.1: Frequency of Price Adjustment by Industry

This table reports average frequencies of price adjustment at the industry and aggregate levels with standard deviations in parentheses for different measures of the frequency of price adjustment. SA treats missing values as interrupting price spells, for SB, missing values do not interrupt price spells if the price is the same before and after periods of missing values and SC forms the union of the two. Columns (1) to (3) use equally weighted frequencies of price adjustments, U, whereas columns (4) to (6) weight frequencies with associated values of shipments, W. Frequencies of price adjustments are calculated at the firm level using the micro data underlying the Producer Price Index constructed by the Bureau of Labor Statistics. The sample period is July 1963 to June 2011

		SAU	SBU	SCU	SAW	SBW	SCW
		(1)	(2)	(3)	(4)	(5)	(6)
Agriculture	Mean	19.07%	21.77%	21.87%	21.43%	24.74%	24.86%
	Std	-16.77%	17.11%	17.20%	18.45%	18.94%	19.05%
	Max	54.24%	64.83%	65.17%	55.83%	64.83%	65.17%
	Min	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
	N		90			86	
Manufacturing	Mean	11.78%	13.00%	13.07%	12.60%	13.87%	13.95%
	Std	-11.35%	11.51%	11.59%	13.00%	13.05%	13.15%
	Max	59.48%	60.32%	60.32%	59.95%	60.32%	60.32%
	Min	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
	N		354			346	
Utilities	Mean	19.74%	21.08%	21.24%	19.96%	21.18%	21.33%
	Std	-13.54%	13.20%	13.22%	13.88%	13.63%	13.66%
	Max	53.89%	53.29%	53.29%	52.63%	52.42%	52.42%
	Min	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
	N		107			105	
Trade	Mean	20.89%	22.98%	23.09%	21.56%	23.76%	23.87%
	Std	-15.54%	15.02%	15.10%	15.41%	14.73%	14.83%
	Max	60.00%	60.00%	60.00%	60.00%	60.00%	60.00%
	Min	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
	N		47			47	
Finance	Mean	13.07%	18.06%	18.18%	13.19%	19.41%	19.56%
	Std	-11.47%	12.63%	12.74%	12.25%	14.47%	14.64%
	Max	45.65%	45.65%	45.65%	46.84%	51.67%	51.67%
	Min	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
	N		122			119	
Service	Mean	9.66%	11.46%	11.50%	10.34%	12.08%	12.12%
	Std	-10.08%	10.90%	10.94%	10.92%	11.56%	11.59%
	Max	43.02%	44.65%	45.12%	43.88%	44.86%	45.28%
	Min	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
	N		72			69	
Total	Mean	14.23%	16.32%	16.41%	15.02%	17.37%	17.48%
	Std	-13.09%	13.43%	13.50%	14.23%	14.72%	14.82%
	Max	60.00%	64.83%	65.17%	60.00%	64.83%	65.17%
	Min	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
	N		792			772	

Table A.2: Summary Statistics and Correlations for Firm Characteristics and Return Predictors (Benchmark Sample, all firms)

This table reports time series averages of annual cross-sectional means and standard deviations for firm characteristics and return predictors used in the subsequent analysis in Panel A and contemporaneous correlations of these variables in Panel B for all firms which have been part of the S&P500 between 1994 and 2009 independent of whether they have missing data on the frequency of price adjustment. Size is the natural logarithm of the market capitalization in thousands, BM is the book to market ratio, Beta is the regression coefficient on the frequency of price adjustment. Size is the natural logarithm of the market capitalization in thousands, CF measures cash flows, Turnover the fraction of shares traded to shares outstanding, spread is the mean bid - ask spread, PCM is the price to cost margin and HHI the Herfindahl - Hirschman index of sales at the Fama & French 48 industry level. Stock level data are from CRSP and financial statement data from Compustat. The sample period is July 1982 to June 2007.

	Size (1)	BM (2)	Beta (3)	Lev (4)	CF (5)	Turnover (6)	Spread (7)	PCM (8)	HHI (9)
Panel A. Means and Standard Deviations									
Mean	14.55	0.63	1.08	0.39	0.09	0.11	0.01	0.37	0.07
Std	1.31	0.39	0.42	0.24	0.07	0.09	0.01	0.19	0.06
N	765	747	737	759	760	765	765	759	752
Panel B. Contemporaneous Correlations									
Size									
BM	-0.18								
Beta	-0.19	-0.14							
Lev	0.01	0.25	-0.11						
CF	0.20	-0.43	-0.10	-0.47					
Turnover	-0.18	-0.11	0.42	-0.17	0.03				
Spread	-0.36	0.15	0.14	0.10	-0.16	-0.01			
PCM	0.13	-0.33	0.06	-0.09	0.28	0.08	-0.14		
HHI	-0.02	-0.14	0.08	-0.07	0.12	0.03	0.06	0.04	

Table A.3: Panel Regressions of Price Stickiness on Firm Characteristics (Benchmark Sample)

This table reports the results of regressing the frequency of price adjustment, SAU, on various firm characteristics and return predictors. Standard errors are clustered at the firms level and reported in parentheses. Equally weighted probabilities of price adjustments are calculated at the firm level using the micro data underlying the Producer Price Index constructed by the Bureau of Labor Statistics. Size is the natural logarithm of the market capitalization in thousands, BM is the book to market ratio, Beta is the regression coefficient on the market excess return in rolling times series regressions, Lev is financial leverage, CF measures cash flows, Turnover the fraction of shares traded to shares outstanding, Spread is the mean bid - ask spread, PCM is the price to cost margin and HHI the Herfindahl - Hirschman index of sales at the Fama & French 48 industry level. Stock level data are from CRSP and financial statement data from Compustat. The sample period is July 1982 to June 2007.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Size	0.46 * *									0.98*** (0.28)
BM		6.91*** (0.79)								7.27*** (0.87)
Beta			-4.88*** (1.13)							-2.86 * * (1.17)
Lev				8.74*** (1.57)						10.41*** (1.68)
CF					-6.62 (5.60)					37.53*** (6.44)
Turnover						-4.25 (3.07)				11.75*** (3.12)
Spread							0.31 (0.21)			0.54 * * (0.26)
PCM								-10.08*** (2.30)		-8.02*** (2.39)
HHI									-26.76*** (5.77)	-18.91*** (5.44)
Observations	14,059	13,823	13,627	13,998	14,009	14,058	14,058	14,007	13,899	13,284
R ²	0.29%	5.11%	2.69%	2.68%	0.10%	0.12%	0.06%	2.19%	1.46%	11.64%

Standard errors in parentheses

*p < 0.10, **p < 0.05, ***p < 0.01

Table A.4: Summary Statistics of Firm Characteristics and Return Predictors at the Portfolio Level (Benchmark Sample)

This table reports time series averages of annual mean firm characteristics and return predictors used in the subsequent analysis at the portfolio level. Stocks are assigned to one of five basket based on the frequency of price adjustment, SAU. Equally weighted probabilities of price adjustments are calculated at the firm level using the micro data underlying the Producer Price Index constructed by the Bureau of Labor Statistics. Size is the natural logarithm of the market capitalization in thousands, BM is the book to market ratio, Beta is the regression coefficient on the market excess return in rolling times series regressions, Lev is financial leverage, CF measures cash flows, Turnover the fraction of shares traded to shares outstanding, Spread is the mean bid - ask spread, PCM is the price to cost margin and HHI the Herfindahl - Hirschman index of sales at the Fama & French 48 industry level. Stock level data are from CRSP and financial statement data from Compustat. The sample period is July 1982 to June 2007.

Portfolio	SAU (1)	Size (2)	BM (3)	Beta (4)	Lev (5)	CF (6)	Turnover (7)	Spread (8)	PCM (9)	HHI (10)
Sticky	0.01	14.47	0.58	1.15	0.38	0.08	0.11	0.01	0.39	0.07
S2	0.05	14.67	0.51	1.09	0.33	0.10	0.10	0.01	0.39	0.07
S3	0.10	14.81	0.59	1.10	0.37	0.09	0.09	0.01	0.37	0.08
S4	0.19	14.86	0.68	1.01	0.45	0.09	0.09	0.01	0.36	0.07
Flexible	0.35	14.84	0.79	0.93	0.45	0.09	0.09	0.01	0.32	0.06
S5-S1	0.34	0.37	0.21	-0.21	0.07	0.00	-0.02	0.00	-0.07	-0.02

Table A.5: Panel Regressions of Annual Stock Returns on Firm Characteristics (Benchmark Sample)

This table reports the results of regressing annual percentage returns on firm characteristics and return predictors. Standard errors are clustered at the firm level and reported in parentheses. Size is the natural logarithm of the market capitalization in thousands, BM is the book to market ratio, Beta is the regression coefficient on the market excess return in rolling times series regressions, Lev is financial leverage, CF measures cash flows, Turnover the fraction of shares traded to shares outstanding, Spread is the mean bid - ask spread, PCM is the price to cost margin and HHI the Herfindahl - Hirschman index of sales at the Fama & French 48 industry level. Stock level data are from CRSP and financial statement data from Compustat. The sample period is July 1982 to June 2007.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Size	-5.03*** (0.25)									-6.48*** (0.32)
BM		8.10*** (0.82)								8.94*** (1.08)
Beta			4.69*** (0.68)							2.00* (1.03)
Lev				-3.07** (1.34)						2.07 (1.98)
CF					-1.25 (5.27)					30.84 (7.86)
Turnover						27.06*** (3.50)				16.43*** (4.65)
Spread							-1.57*** (0.35)			-5.93*** (0.44)
PCM								2.31 (1.56)		10.08*** (2.18)
HHI									16.08*** (5.12)	15.22** (6.42)
Year Fixed Effects	N	N	N	N	N	N	N	N	N	N
Observations	14,026	13,791	13,594	13,965	13,976	14,026	14,026	13,974	13,866	13,253
R ²	3.82%	0.77%	0.26%	0.04%	0.00%	0.51%	0.17%	0.01%	0.06%	24.78%

Standard errors in parentheses
 * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.6: CAPM Regressions (Benchmark Sample)

This table reports results for the unconditional CAPM. Stocks are assigned to one of five baskets based on the frequency of price adjustment, SAU and returns are equally weighted at the portfolio level. Equally weighted probabilities of price adjustments are calculated at the firm level using the micro data underlying the Producer Price Index constructed by the Bureau of Labor Statistics. α is the intercept and β the slope of times series regressions of monthly portfolio excess returns on a constant and the excess return of the CRSP value weighted index. OLS and Fama and MacBeth (1973) standard errors are reported in parentheses and Newey and West (1987) standard errors in brackets. The sample period is July 1982 to June 2007.

	Sticky	S2	S3	S4	Flexible	S1-S5
α_p	0.56	0.49	0.49	0.48	0.47	0.10
SE_{OLS}	(0.10)***	(0.09)***	(0.10)***	(0.10)***	(0.13)***	(0.12)***
SE_{NW}	[0.19]***	[0.13]***	[0.18]***	[0.15]***	[0.18]***	[0.13]***
β_p	1.13	1.08	1.04	0.96	0.87	0.26
SE_{OLS}	(0.02)***	(0.02)***	(0.02)***	(0.02)***	(0.03)***	(0.03)***
SE_{NW}	[0.04]***	[0.04]***	[0.06]***	[0.05]***	[0.07]***	[0.04]***

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.7: Long Horizon Predictability: HML and Market (Benchmark Sample)

This table reports results for m-month forecasting regressions of log excess returns on the zero-cost portfolio of going long the portfolio of stocks with high book to market values of equity and shorting the portfolio of stocks with low book to market values of equity of Fama and French (1993), HML, in Panel A, and the CRSP value weighted excess return in Panel B on the proxy for the consumption wealth ratio of Lettau and Ludvigson (2001). For each regression the table reports OLS standard errors in parentheses, Newey and West (1987) standard errors in brackets, and Hodrick (1992) standard errors in curly brackets. The sample period is July 1982 to June 2007.

Horizon m (Months)	1	6	12	24	36	48	60
Panel A. HML							
$b_{lh}^{(m)}$	-0.13 (0.10)	-0.90 (0.29)***	-1.85 (0.46)***	-4.17 (0.71)***	-6.49 (0.78)***	-7.40 (0.89)***	-8.41 (1.03)***
SE_{OLS}							
SE_{NW}	[0.07]*	[0.35]***	[0.83]**	[1.51]***	[2.38]***	[3.90]*	[4.46]*
SE_H	{0.07}*	{0.44}**	{0.94}**	{2.08}**	{3.54}*	{5.06}	{5.21}
R^2	0.57%	3.18%	5.38%	11.25%	20.84%	21.59%	21.96%
Panel B. Market Excess Return							
$b_{lh}^{(m)}$	0.05 (0.14)	0.61 (0.36)*	1.68 (0.52)***	4.77 (0.78)***	9.68 (1.04)***	14.30 (1.23)***	14.69 (1.38)***
SE_{OLS}							
SE_{NW}	[0.10]	[0.49]	[1.01]*	[2.64]*	[4.70]**	[5.11]***	[5.01]***
SE_H	{0.10}	{0.61}	{1.27}	{2.80}*	{4.97}*	{8.25}*	{9.10}
R^2	0.04%	1.00%	3.52%	11.93%	25.00%	34.94%	32.24%

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.8: **Calibration**

This table shows in Panel A calibrated parameter values of the model of Section IV and the sectoral distribution of the frequency of price adjustment in Panel B.

Panel A. Calibration Parameter		
Parameter	Value	Source
β	0.99	standard
b	0.76	Altig et al. (2011)
γ	5	Jermann (1998)
σ	2.5	Carvalho (2006)
ψ	1	Altig et al. (2011)
ϵ_c	8	Carvalho (2006)
ϵ_{ck}	12	Carvalho (2006)
θ_w	0.825	Erceg et al. (2000)
ϵ_w	8	Altig et al. (2011) / Erceg et al. (2000)
ϕ_π	1.24	Rudebusch (2002)
ϕ_y	0.33/4	Rudebusch (2002)
ρ_a	0.95	Smets and Wouters (2007)
ρ_m	0.90	Coibion and Gorodnichenko (2012)
Panel B. Sectoral Distribution		
Sector k	Share	Frequency of Price Adjustment
1	0.2	0.105
2	0.2	0.164
3	0.2	0.277
4	0.2	0.638
5	0.2	0.985

Table A.9: Long-Horizon Predictability on Simulated Data

This table reports results for q-quarters forecasting regressions of model implied log excess returns of the zero-cost portfolio of going long the claim to dividends of the sector with low frequencies of price adjustment and shorting the claim to dividends of the sector with high frequencies of price adjustment, L-H on log consumption surplus of the model. For each regression the table reports OLS standard errors in parentheses, Newey and West (1987) standard errors in brackets, Hansen and Hodrick (1980) standard errors in curly brackets and Hodrick (1992) standard errors in angle brackets. The sample length is 250 quarters.

Horizon q (Quarters)	1	2	4	8	12	16	20
$b_{lh}^{(q)}$	-0.14	-0.26	-0.48	-0.79	-0.81	-0.77	-0.68
SE_{OLS}	(0.03)***	(0.04)***	(0.06)***	(0.10)***	(0.13)***	(0.17)***	(0.20)***
SE_{NW}	[0.05]***	[0.07]***	[0.11]***	[0.20]***	[0.29]***	[0.40]*	[0.50]
SE_{HH}	{0.05}***	{0.08}***	{0.13}***	{0.23}***	{0.31}***	{0.43}*	{0.52}
SE_H	(0.05)***	(0.10)***	(0.17)***	(0.26)***	(0.32)**	(0.36)**	(0.38)*
R^2	9.34%	14.28%	21.86%	21.80%	13.75%	8.61%	5.05%

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.10: Panel Regression of Annual Stock Returns on Price Stickiness and Firm Characteristics (Benchmark Sample, Monthly Overlapping)

This table reports the results of regressing overlapping annual percentage returns at the monthly frequency on the frequency of price adjustment, SAU, firm characteristics, return predictors and year fixed effects, where indicated. Standard errors are clustered at the firm level and reported in parentheses. Equally weighted probabilities of price adjustments are calculated at the firm level using the micro data underlying the Producer Price Index constructed by the Bureau of Labor Statistics. Size is the natural logarithm of the market capitalization in thousands, BM is the book to market ratio, Beta is the regression coefficient on the market excess return in rolling times series regressions, Lev is financial leverage, CF measures cash flows, Turnover the fraction of shares traded to shares outstanding, Spread is the mean bid - ask spread, PCM is the price to cost margin and HHI the Herfindahl - Hirschman index of sales at the Fama & French 48 industry level. Stock level data are from CRSP and financial statement data from Compustat. The sample period is July 1982 to June 2007.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
SAU	-7.88*** (1.68)	-6.14*** (1.78)	-12.49*** (2.03)	-5.23*** (1.67)	-7.93*** (1.73)	-7.99*** (1.71)	-8.35*** (1.69)	-7.56*** (1.69)	-7.93*** (1.73)	-7.35*** (1.69)	-4.66 * * (2.15)
Size		-2.72*** (0.12)									-4.22*** (0.21)
BM			6.70*** (0.53)								6.23*** (0.69)
Beta				4.89*** (0.52)							5.95*** (0.75)
Lev					0.87 (0.96)						1.99 (1.31)
CF						-9.37 * * (4.00)					18.13*** (5.81)
Turnover							-2.03 (2.23)				-10.74*** (2.87)
Spread								-1.75*** (0.16)			-5.74*** (0.25)
PCM									-0.70 (1.28)		7.45*** (1.66)
HHI										10.32*** (3.41)	6.37 (4.51)
Observations	264,662	264,662	260,096	256,393	261,913	262,757	256,941	264,657	262,636	247,781	239,309
R ²	0.06%	1.84%	0.83%	0.63%	0.06%	0.08%	0.07%	0.43%	0.06%	0.08%	4.81%
Observations	285,503	285,503	280,335	277,689	282,834	283,786	277,572	285,498	283,654	280,179	262,864
R ²	0.07%	1.68%	0.81%	0.33%	0.07%	0.09%	0.08%	0.41%	0.07%	0.10%	4.51%

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.11: Panel Regressions of Annual Stock Returns on Price Stickiness and Firm Characteristics (Benchmark Sample, Durability)

This table reports the results of regressing annual percentage returns on the frequency of price adjustment, SAU, firm characteristics, return predictors and year fixed effects, where indicated. Standard errors are clustered at the firm level and reported in parentheses. Equally weighted probabilities of price adjustments are calculated at the firm level using the micro data underlying the Producer Price Index constructed by the Bureau of Labor Statistics. Size is the natural logarithm of the market capitalization in thousands, BM is the book to market ratio, Beta is the regression coefficient on the market excess return in rolling times series regressions, Lev is financial leverage, CF measures cash flows, Turnover the fraction of shares traded to shares outstanding, Spread is the mean bid - ask spread, PCM is the price to cost margin, HHI the Herfindahl - Hirschman index of sales at the Fama & French 48 industry level and Dur the durability measure of Bils et al. (2012). Stock level data are from CRSP and financial statement data from Compustat. The sample period is July 1982 to June 2007.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)
SAU	-10.18*** (2.14)	-11.07*** (2.25)	-7.98*** (2.42)	-12.97*** (2.47)	-7.87*** (2.30)	-11.01*** (2.29)	-10.98*** (2.28)	-9.87*** (2.24)	24.65*** (1.06)	-9.53*** (2.22)	-10.74*** (2.27)	-11.52*** (2.47)	-8.57*** (2.96)
Size			-4.34*** (0.28)										-5.00*** (0.34)
BM				3.31*** (0.83)									3.97*** (1.18)
Beta					5.30*** (0.72)								2.74*** (1.06)
Lev						1.02 (1.35)							4.60** (1.88)
CF							-10.56* (5.49)						13.40 (8.30)
Turnover								52.49*** (3.82)					33.72*** (5.02)
Spread									-0.10*** (0.02)				-7.99*** (0.61)
PCM										5.87*** (1.63)			6.28*** (2.17)
HHI											2.95 (4.70)		11.29* (6.41)
Dur												0.08 (0.32)	-1.42*** (0.41)
Year Fixed Effects	N	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Observations	14,026	14,026	14,026	13,791	13,594	13,965	13,976	14,026	14,026	13,974	13,866	12,236	11,770
R ²	0.11%	19.83%	21.85%	19.94%	20.36%	19.72%	19.74%	21.35%	20.91%	19.80%	19.41%	18.75%	23.99%

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.12: Panel Regressions of Annual Stock Returns on Price Stickiness and Firm Characteristics (Benchmark Sample, Double Clustered)

This table reports the results of regressing annual percentage returns on the frequency of price adjustment, SAU, firm characteristics, return predictors, and year fixed effects, where indicated. SStandard errors are clustered at the firm level or firm-year level and reported in parentheses. Equally-weighted probabilities of price adjustments are calculated at the firm level using the micro data underlying the Producer Price Index constructed by the Bureau of Labor Statistics. Size is the natural logarithm of the market capitalization in thousands, BM is the book-to-market ratio, Beta is the regression coefficient on the market excess return in rolling times-series regressions, Lev is financial leverage, CF measures cash flows, Turnover is the fraction of shares traded to shares outstanding, Spread is the mean bid - ask spread, PCM is the price-to-cost margin, and HHI is the Herfindahl - Hirschman index of sales at the Fama & French 48 industry level. Stock-level data are from CRSP and financial statement data are from Compustat. The sample period is July 1982 to June 2007.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)
SAU	-10.18*** (1.06)	-11.07*** (2.25)	-7.98*** (2.42)	-12.97*** (2.47)	-7.87*** (2.30)	-11.01*** (2.29)	-10.98*** (2.28)	-9.87*** (2.24)	-10.31*** (2.32)	-9.53*** (2.22)	-10.74*** (2.27)	-6.46 * * (2.69)	-6.46* (3.49)
Size			-4.34*** (0.28)									-4.87*** (0.31)	-4.87*** (1.01)
BM				3.31*** (0.83)								3.88*** (1.07)	3.88 (2.74)
Beta					5.30*** (0.72)							1.23 (0.91)	1.23 (3.74)
Lev						1.02 (1.35)						4.80*** (1.78)	4.80 (3.36)
CF							-10.56* (5.49)					5.67 (7.51)	5.67 (11.39)
Turnover								52.49*** (3.82)				35.06*** (4.80)	35.06* (18.89)
Spread									-5.55*** (0.54)			-7.60*** (0.57)	-7.60*** (1.27)
PCM										5.87*** (1.63)		8.00*** (1.96)	8.00* (4.12)
HHI											2.95 (4.70)	11.28 * * (5.54)	11.28 (10.16)
Year Fixed Effects	N	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Cluster	Firm	Firm	Firm	Firm	Firm	Firm	Firm	Firm	Firm	Firm	Firm	Firm	Firm-Year
Observations	14,026	14,026	14,026	13,791	13,594	13,965	13,976	14,026	14,026	13,974	13,866	13,253	13,253
R ²	0.11%	19.83%	21.85%	19.94%	20.36%	19.72%	19.74%	21.35%	20.91%	19.80%	19.41%	24.82%	24.82%

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.13: Panel Regressions of Annual Stock Returns on Price Stickiness and Firm Characteristics (Benchmark Sample, Within Industry)

This table reports the results of regressing annual percentage returns on the frequency of price adjustment, SAU, firm characteristics, return predictors, year fixed effects and industry fixed effects, where indicated. FF10 and FF17 indicate industry dummies according to the Fama & French 10 and 17 industry definition. Standard errors are clustered at the firm level and reported in parentheses. Equally weighted probabilities of price adjustments are calculated at the firm level using the micro data underlying the Producer Price Index constructed by the Bureau of Labor Statistics. The sample period is July 1982 to June 2007.

	Baseline (1)	Agriculture (2)	Manufacturing (3)	Utilities (4)	Trade (5)	Finance (6)	Services (7)	Dummies (8)	Dummies (9)	Dummies (10)
SAU	-11.07*** (2.25)	-15.4*** (6.85)	-7.38* (4.08)	-9.73** (4.14)	-9.13 (8.92)	-2.50 (4.78)	-12.16 (19.37)	-7.93*** (2.40)	-5.73** (2.59)	-4.66* (2.53)
Year Fixed Effects	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Industry Fixed Effects	N	N	N	N	N	N	N	Y	FF10	FF17
Observations	14,026	764	6,932	2,075	1,058	2,270	927	14,026	14,008	14,008
R ²	19.83%	26.88%	20.24%	25.34%	39.26%	44.46%	21.62%	20.10%	20.09%	19.94%

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.14: Panel Regressions of Annual Stock Returns on Price Stickiness and Firm Characteristics (Benchmark Sample, Unwinsorized)

This table reports the results of regressing annual percentage returns on the frequency of price adjustment, SAU, firm characteristics, return predictors and year fixed effects, where indicated. Standard errors are clustered at the firm level and reported in parentheses. Equally weighted probabilities of price adjustments are calculated at the firm level using the micro data underlying the Producer Price Index constructed by the Bureau of Labor Statistics. Size is the natural logarithm of the market capitalization in thousands, BM is the book to market ratio, Beta is the regression coefficient on the market excess return in rolling times series regressions, Lev is financial leverage, CF measures cash flows, Turnover the fraction of shares traded to shares outstanding, Spread is the mean bid - ask spread, PCM is the price to cost margin and HHI the Herfindahl - Hirschman index of sales at the Fama & French 48 industry level. Stock level data are from CRSP and financial statement data from Compustat. The sample period is July 1982 to June 2007.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
SAU	-11.36*** (2.87)	-12.26*** (3.00)	-8.63*** (3.14)	-13.17*** (3.16)	-6.86** (2.98)	-12.56*** (3.01)	-12.06*** (3.00)	-10.83*** (2.85)	-11.76*** (3.05)	-11.72*** (2.94)	-11.57*** (2.98)	-6.23* (3.30)
Size			-5.70*** (0.45)									-5.77*** (0.45)
BM				1.90** (0.97)								1.86* (1.12)
Beta					9.04*** (1.08)							1.94 (1.24)
Lev						2.88 (1.98)						4.17* (2.29)
CF							-23.31* (14.17)					0.95 (9.84)
Turnover								61.87*** (7.03)				45.46*** (7.76)
Spread									-2.19*** (0.84)			-4.97*** (1.12)
PCM										0.69 (0.67)		3.02*** (0.75)
HHI											1.22 (4.62)	7.23 (4.62)
Year Fixed Effects	N	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Observations	14,026	14,026	14,026	13,791	13,594	13,965	13,976	14,026	14,026	13,974	13,866	13,253
R ²	0.10%	13.85%	16.44%	14.44%	14.67%	13.77%	13.90%	16.24%	14.26%	13.74%	13.47%	19.92%

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.15: Mean Portfolio Returns (Alternative Measures of Price Stickiness)

This table reports time series averages of annual equally weighted portfolio raw returns for various sample periods with Newey and West (1987) standard errors in parentheses. Stocks are assigned to one of five baskets based on the frequency of price adjustment. SA treats missing values as interrupting price spells, for SB, missing values do not interrupt price spells if the price is the same before and after periods of missing values and SC forms the union of the two. Equally (U) and value of shipment weighted (W) probabilities of price adjustments are calculated at the firm level using the micro data underlying the Producer Price Index constructed by the Bureau of Labor Statistics.

	Sticky (1)	S2 (2)	S3 (3)	S4 (4)	Flexible (5)	S1-S5 (6)
Panel A. Annual Mean Returns (SBU)						
07/1982 - 06/2014	21.56*** (3.75)	19.8***1 (2.81)	19.77*** (2.88)	18.97*** (3.23)	18.49*** (2.52)	3.06* (1.72)
07/1982 - 06/2007	24.05*** (3.00)	21.48*** (2.50)	21.65*** (2.49)	21.50*** (2.46)	20.46*** (2.41)	3.59 * * (1.78)
Panel B. Annual Mean Returns (SCU)						
07/1982 - 06/2014	21.53*** (3.76)	19.79*** (2.78)	19.80*** (2.90)	18.97*** (3.23)	18.49*** (2.52)	3.04* (1.71)
07/1982 - 06/2007	24.04*** (3.03)	21.43*** (2.46)	21.66*** (2.54)	21.53*** (2.42)	20.46*** (2.41)	3.58 * * (1.77)
Panel C. Annual Mean Returns (SAW)						
07/1982 - 06/2014	21.15*** (3.36)	19.91*** (3.04)	19.97*** (2.99)	18.88*** (3.08)	18.06*** (2.47)	3.08* (1.78)
07/1982 - 06/2007	23.54*** (2.52)	21.58*** (2.81)	21.95*** (2.50)	21.35*** (2.29)	19.75*** (2.60)	3.78 * * (1.73)
Panel D. Annual Mean Returns (SBW)						
07/1982 - 06/2014	21.64*** (3.45)	18.78*** (2.79)	19.93*** (2.80)	18.78*** (3.32)	18.76*** (2.55)	2.89* (1.74)
07/1982 - 06/2007	24.01*** (2.67)	20.24*** (2.58)	21.80*** (2.42)	21.39*** (2.41)	20.63*** (2.65)	3.38* (1.85)
Panel E. Annual Mean Returns (SCW)						
07/1982 - 06/2014	21.62*** (3.46)	18.85*** (2.81)	19.93*** (2.80)	18.79*** (3.37)	18.75*** (2.52)	2.87* (1.70)
07/1982 - 06/2007	24.05*** (2.64)	20.23*** (2.62)	21.80*** (2.42)	21.44*** (2.44)	20.57*** (2.65)	3.47* (1.92)

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.16: Mean Portfolio Returns (SAU, S&P500)

This table reports time series averages of annual equally weighted portfolio raw returns for various sample periods with Newey and West (1987) standard errors in parentheses. Stocks are assigned to one of five baskets based on the frequency of price adjustment, SAU. Equally weighted probabilities of price adjustments are calculated at the firm level using the micro data underlying the Producer Price Index constructed by the Bureau of Labor Statistics. Panel A includes only firms which have been part of the S&P500 index at the end of June 1994 whereas Panel B focuses on firms which have been part of the S&P500 index at the end of June 2006.

	Sticky (1)	S2 (2)	S3 (3)	S4 (4)	Flexible (5)	S1-S5 (6)
Panel A. S&P500 Constituents as of 06/30/1994						
07/1982 - 06/2014	18.41*** (3.54)	18.01*** (3.29)	18.36*** (2.83)	17.06*** (3.09)	15.71*** (2.65)	2.70* (1.58)
07/1982 - 06/2007	20.30*** (3.33)	19.39*** (3.16)	19.87*** (2.88)	18.86*** (2.99)	17.26*** (3.12)	3.04 * * (1.38)
Panel B. S&P500 Constituents as of 06/30/2006						
07/1982 - 06/2014	22.94*** (4.38)	20.51*** (3.46)	20.26*** (3.21)	19.39*** (3.86)	18.22*** (2.63)	4.72 * * (1.97)
07/1982 - 06/2007	25.67*** (3.56)	22.57*** (3.01)	22.44*** (2.41)	22.32*** (2.70)	20.50*** (2.15)	5.17 * * (2.12)

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.17: Panel Regressions of Annual Realized Volatilities on Price Stickiness and Firm Characteristics (Benchmark Sample)

This table reports the results of regressing annual realized volatilities on the frequency of price adjustment, SAU, firm characteristics, return predictors and year fixed effects, where indicated. Standard errors are clustered at the firm level and reported in parentheses. Equally weighted probabilities of price adjustments are calculated at the firm level using the micro data underlying the Producer Price Index constructed by the Bureau of Labor Statistics. Realized volatilities are calculated as the square root of the sum of daily squared returns. Size is the natural logarithm of the market capitalization in thousands, BM is the book to market ratio, Beta is the regression coefficient on the market excess return in rolling times series regressions, Lev is financial leverage, CF measures cash flows, Turnover the fraction of shares traded to shares outstanding, Spread is the mean bid - ask spread, PCM is the price to cost margin and HHI the Herfindahl - Hirschman index of sales at the Fama & French 48 industry level. Stock level data are from CRSP and financial statement data from Compustat. The sample period is July 1982 to June 2007.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
SAU	-10.08*** (3.22)	-9.94*** (3.16)	-7.41** (2.91)	-7.72** (3.10)	-0.56 (2.29)	-7.01** (3.07)	-9.84*** (3.18)	-8.00*** (2.17)	-10.82*** (2.88)	-8.73*** (3.12)	-8.76*** (3.15)	-6.46** (2.69)
Size			-3.55*** (0.27)									-4.87*** (0.31)
BM				-3.70*** (0.88)								3.88*** (1.07)
Beta					18.53*** (0.78)							1.23 (0.91)
Lev						-8.71*** (1.54)						4.80*** (1.78)
CF							-9.80* (5.67)					5.67 (7.51)
Turnover								84.73*** (2.65)				35.06*** (4.80)
Spread									6.28*** (0.52)			-7.60*** (0.57)
PCM										4.42** (2.11)		8.00*** (1.96)
HHI											12.39** (5.14)	11.28** (5.54)
Year Fixed Effects	N	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Observations	14,057	14,057	14,057	13,821	13,625	13,996	14,007	14,057	14,057	14,005	13,897	13,253
R ²	0.59%	22.94%	30.25%	24.35%	45.51%	24.57%	23.15%	44.40%	30.44%	23.26%	23.32%	24.82%

Standard errors in parentheses

*p < 0.10, **p < 0.05, ***p < 0.01

Table A.18: Summary Statistics and Correlations for Firm Characteristics and Return Predictors (Full Sample)

This table reports time series averages of annual cross-sectional means and standard deviations for firm characteristics and return predictors used in the subsequent analysis in Panel A and contemporaneous correlations of these variables in Panel B. SAU measures the frequency of price adjustment. Equally weighted probabilities of price adjustments are calculated at the firm level using the micro data underlying the Producer Price Index constructed by the Bureau of Labor Statistics. Size is the natural logarithm of the market capitalization in thousands, BM is the book to market ratio, Beta is the regression coefficient on the market excess return in rolling times series regressions, Lev is financial leverage, CF measures cash flows, Turnover the fraction of shares traded to shares outstanding, Spread is the mean bid - ask spread, PCM is the price to cost margin and HHI the Herfindahl - Hirschman index of sales at the Fama & French 48 industry level. Stock level data are from CRSP and financial statement data from Compustat. The sample period is July 1982 to June 2014.

	SAU (1)	Size (2)	BM (3)	Beta (4)	Lev (5)	CF (6)	Turnover (7)	Spread (8)	PCM (9)	HHI (10)
Panel A. Means and Standard Deviations										
Mean	0.14	15.03	0.63	1.06	0.40	0.09	0.14	0.01	0.37	0.07
Std	0.13	1.25	0.38	0.43	0.24	0.06	0.10	0.01	0.19	0.06
N	540	540	530	523	538	539	540	540	538	535
Panel B. Contemporaneous Correlations										
Size	0.08									
BM	0.24	-0.17								
Beta	-0.16	-0.21	-0.12							
Lev	0.15	-0.01	0.20	-0.13						
CF	-0.03	0.20	-0.45	-0.08	-0.45					
Turnover	0.00	-0.22	-0.05	0.44	-0.11	0.02				
Spread	0.02	-0.35	0.18	0.18	0.08	-0.18	0.06			
PCM	-0.15	0.13	-0.32	0.06	-0.09	0.29	0.07	-0.12		
HHI	-0.13	0.00	-0.20	0.09	-0.06	0.15	0.01	0.02	0.03	

Table A.19: Panel Regressions of Annual Stock Returns on Price Stickiness and Firm Characteristics (Full Sample)

This table reports the results of regressing annual percentage returns on the frequency of price adjustment, SAU, firm characteristics, return predictors and year fixed effects, where indicated. Standard errors are clustered at the firm level and reported in parentheses. SAU treats missing values as interrupting price spells. Equally-weighted probabilities of price adjustments are calculated at the firm level using the micro data underlying the Producer Price Index constructed by the Bureau of Labor Statistics. Size is the natural logarithm of the market capitalization in thousands, BM is the book to market ratio, Beta is regression the coefficient on the market excess return in rolling times series regressions, Lev is financial leverage, CF measures cash flows, Turnover the fraction of shares traded to shares outstanding, Spread is the mean bid - ask spread, PCM is the price to cost margin and HHI the Herfindahl - Hirschman index of sales at the Fama & French 48 industry level. Stock level data are from CRSP and financial statement data from Compustat. The sample period is July 1982 to June 2014.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
SAU	-8.11*** (1.88)	-8.97*** (2.02)	-6.12*** (2.17)	-10.91*** (2.18)	-6.86*** (2.07)	-9.00*** (2.06)	-8.88*** (2.05)	-9.09*** (2.00)	-8.34*** (2.07)	-7.88*** (2.00)	-8.49*** (2.03)	-4.50* (2.39)
Size			-3.90*** (0.25)									-4.66*** (0.27)
BM				3.32*** (0.70)								3.21*** (0.89)
Beta					3.74*** (0.60)							1.87 ** (0.76)
Lev						1.12 (1.16)						3.42 ** (1.55)
CF							-8.09 ** (4.59)					9.53 (6.56)
Turnover								22.95*** (3.02)				7.22 ** (3.51)
Spread									-5.67*** (0.54)			-7.86*** (0.56)
PCM										4.01*** (1.38)		7.59*** (1.71)
HHI											5.84 (4.16)	11.32 ** (5.07)
Year Fixed Effects	N	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Observations	17,243	17,243	17,243	16,926	16,691	17,176	17,192	17,243	17,243	17,187	17,069	16,251
R ²	0.07%	24.10%	25.74%	24.21%	24.39%	24.03%	24.04%	24.54%	25.04%	24.07%	23.82%	27.69%

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$